Two-Stage Calibration of Decomposable Web Ranking: Block Escape Probability, Incoming-Transition Deviation, and the Anchoring of Oracle Damping

Ioannis Konstantellos
Computer Engineering and Informatics Department, University of Patras, Greece
st1059599@ceid.upatras.gr, idellos1999@gmail.com

October 2026

Abstract

NCDawareRank combines the hyperlink matrix H of a Web graph with an inter-level proximity matrix M defined over nearly completely decomposable (NCD) blocks, and its two weights η and μ are fixed by convention. Writing $\eta = (1 - r)\alpha$ and $\mu = r\alpha$, we formulate and test a two-stage, zero-cost calibration: Stage 1 chooses the mixing ratio r from the mean block escape probability $\bar{P}_{\text{esc}}^{(M)} = \frac{1}{n}\sum_u k_u/(k_u + 1)$, and Stage 2 chooses the total damping α from the deviation h of the ensembled operator $Z_r = (1 - r)H + rM$ from double stochasticity. We prove that $h = \|u^\top Z_r - u^\top\|_1$, that its teleportation-weighted counterpart $h_v = \|v^\top Z_r - v^\top\|_1$ equals the L_1 norm of $\mathrm{d}\pi/\mathrm{d}\alpha$ at $\alpha = 0$, and that $\|\pi(\alpha) - v\|_1 \le \frac{\alpha}{1-\alpha} h_v$ for every α. We evaluate the procedure on 3 007 perturbed instances of synthetic block webs and five real networks, including cnr-2000, one of the Web graphs of the original NCDawareRank experiments. The perturbations follow the original protocol: links removed at random, and newly added pages that have lost most of their in-links. Three findings emerge. First, neither $\bar{P}_{\text{esc}}^{(M)}$ nor h nor h_v predicts the oracle parameters out of sample: inside a graph they track the amount of link loss rather than structure, across graphs their relation with the oracle is weak or reversed, and h moves in opposite directions under the two perturbations. The rules that maximise rank agreement therefore collapse to the constants $r \approx 0.18$–0.24 and $\alpha = 0.810$. Second, these constants recover the canonical ranking better than the static NCDawareRank setting on held-out synthetic graphs (Kendall τ +0.0110, better on 31/32 graphs), but not on the real networks (+0.0017, $p = 0.812$). Third, the oracle damping is set by the reference ranking itself: for references computed on the complete graph it lies 0.03–0.06 below the damping that defines the reference (a noise discount), and it moves with the prior mean of Random-α references with slope 0.950. Whether a calibrated setting beats the static one therefore depends on the ground truth chosen. We also show that the stability protocol commonly used to compare link-analysis methods is degenerate as a calibration target.

1 Introduction

Link analysis in the PageRank tradition [PBMW99, LM06] ranks the pages of a directed graph $\mathcal{G} = (\mathcal{W}, \mathcal{E})$ by the stationary distribution of a random surfer who follows a hyperlink with probability α and teleports otherwise. The Web is also organised hierarchically into sites and domains, which makes its link matrix *nearly completely decomposable* (NCD) in the sense of Simon and Ando [SA61] and Courtois [Cou77]: links are dense inside blocks and sparse between them. NCDawareRank [NG13] exploits this structure by adding to the hyperlink matrix H an *inter-level proximity* matrix M that spreads rank uniformly over the blocks a page touches, giving the chain

$$P = \eta H + \mu M + (1 - \eta - \mu)E, \tag{1}$$

where E is a block-aware teleportation matrix. The method reduces the sensitivity of the ranking to sparse or missing links and to newly added pages, and it accelerates convergence [NG13, Nik20].

Its two weights are chosen by convention. In the original experiments $\eta = 0.85$ and $\mu = 0.10$, so the total damping is $\eta + \mu = 0.95$. A second configuration, $\eta = 0.75$ and $\mu = 0.10$ (total damping 0.85), appears in the convergence comparison. The value $\mu \approx 0.10$ was motivated by iteration counts on complete graphs together

with "the fact that we want to avoid ignoring the direct link structure" [NG13, §3.2.2]. The choice of the PageRank damping factor has a literature of its own: for real Web graphs, values close to 1 do not give more meaningful rankings [BSV05, BSV09]; structural arguments favour values near $1/2$ [ALP08]; and empirically measured teleportation behaviour follows a Beta distribution with mean between 0.3 and 0.7, depending on the site [GCFG10]. Gleich et al. conjectured that good choices of α "may depend on a topological property of the graph" [GCFG10]. In earlier work on uncertain graphs, we calibrated the damping of a diffusion heuristic from a free structural statistic, the mean escape probability [Kon26].

This paper asks the analogous question for NCDawareRank: *can* (η, μ) *be calibrated at no cost from statistics that are already computed when the operator is built?* We write $\eta = (1 - r)\alpha$ and $\mu = r\alpha$, which separates the mixing ratio r between H and M from the total damping α, and we study a sequential two-stage scheme. Stage 1 fixes $\alpha_0 = 0.85$ and sets $r = f(\bar{P}^{(M)}_{\text{esc}})$ from the mean *block escape probability*. Stage 2 sets $\alpha = g(h)$ from the mean absolute deviation h of the incoming transition sums of Z_r from unity.

Contributions.

1. **Framework and theory** (Sections 3–5). We define the two statistics and show that both are computable in $O(|\mathcal{E}| + n)$ time. We prove that $h = \|u^\top Z_r - u^\top\|_1$ and that the teleportation-weighted $h_v = \|v^\top Z_r - v^\top\|_1$ is the L_1 norm of the first derivative of the ranking at $\alpha = 0$. We then prove a displacement bound, $\|\pi(\alpha) - v\|_1 \le \frac{\alpha}{1-\alpha} h_v$ (Theorem 1), and characterise when the ranking equals the teleportation vector. Two statements in the literature are corrected along the way: $h = 0$ does not imply a uniform ranking under block-even teleportation, and singleton blocks turn M into the lazy hyperlink walk rather than into H.
2. **A large, cluster-aware evaluation** (Section 6). We follow the perturbation protocol of [NG13] (random link removal and newly added pages) on 3 007 instances. They come from synthetic block webs with held-out, scale-transfer and topology-transfer panels, and from five real networks. Rankings are scored against seven ground truths, and every test treats the base graph as the unit of replication.
3. **Findings** (Sections 7–8). (i) $\bar{P}^{(M)}_{\text{esc}}$, h and h_v do not predict the oracle parameters, and we explain why. (ii) The calibration reduces to two constants that help on synthetic graphs but not on real ones. (iii) Oracle damping is anchored to the damping that defines the reference ranking, minus a noise discount, as a dose-response experiment shows directly. (iv) The stability protocol of [NG13] is degenerate as a calibration target.

We report the negative results with the same care as the positive ones, because they determine what can and cannot be claimed about calibrating link-analysis parameters from structure.

2 Background

2.1 NCDawareRank

Let $\mathcal{W} = \{1, \dots, n\}$ be the pages, $\mathcal{G}_u$ the set of pages u links to and $d_u = |\mathcal{G}_u|$. The hyperlink matrix has $H_{uv} = 1/d_u$ for $v \in \mathcal{G}_u$; rows of dangling pages ($d_u = 0$) are replaced by $\frac{1}{n}e^\top$, the uniform stochasticity adjustment of [PBMW99]. The original paper leaves the adjustment open [NG13], and the extended version proposes a block-confined alternative [Nik20]. A partition $\mathcal{A} = \{\mathcal{A}_1, \dots, \mathcal{A}_N\}$ of $\mathcal{W}$ into NCD blocks is given, and $\mathcal{A}_{(u)}$ denotes the block of u. Following [NG13, §3.1], the proximal set of u is

$$\mathcal{X}_u = \bigcup_{w \in \{u\} \cup \mathcal{G}_u} \mathcal{A}_{(w)},$$

and N_u is the number of distinct blocks in $\mathcal{X}_u$. The inter-level proximity matrix is

$$M_{uv} = \begin{cases} \dfrac{1}{N_u\,|\mathcal{A}_{(v)}|}, & v \in \mathcal{X}_u, \\ 0, & \text{otherwise}, \end{cases} \qquad M = RA, \tag{2}$$

where $R_{uI} = 1/(N_u|\mathcal{A}_I|)$ if $\mathcal{A}_I \subseteq \mathcal{X}_u$ (and 0 otherwise) and $A_{Iv} = \mathbf{1}[v \in \mathcal{A}_I]$. Thus M is a two-step walk: first choose one of the N_u proximal blocks uniformly, then choose a page inside it uniformly. The factorisation

$M = RA$ keeps the storage at $O(\mathrm{nnz}(R) + n)$. Teleportation is block-even, $E = ev^\top$ with $v_u = 1/(N|\mathcal{A}_{(u)}|)$: rank is spread first evenly over the blocks and then evenly over the pages of each block. The chain is (1) with $\eta > 0, \mu \geq 0$ and $\eta + \mu < 1$. The NCDawareRank vector π satisfies $\pi^\top = \pi^\top P$ and is computed by a factorised power iteration [NG13, Alg. 1].

2.2 The (α, r) parameterisation

In the proof of Theorem 1 of [NG13], the chain is rewritten as $P = \alpha Z + (1 - \alpha)E$ with $\alpha = \eta + \mu$ and $Z = (\eta H + \mu M)/(\eta + \mu)$, which shows that the subdominant eigenvalue of P is at most $\eta + \mu$. We adopt this decomposition and introduce the *mixing ratio* $r = \mu/\alpha \in [0, 1]$:

$$\eta = (1 - r)\alpha, \qquad \mu = r\alpha, \qquad Z_r = (1 - r)H + rM, \qquad P = \alpha Z_r + (1 - \alpha)ev^\top. \tag{3}$$

The ranking is $\pi(\alpha)^\top = (1 - \alpha)v^\top(I - \alpha Z_r)^{-1} = (1 - \alpha)\sum_{k\geq 0} \alpha^k v^\top Z_r^k$. The two published configurations correspond to $(\alpha, r) = (0.85,\ 0.10/0.85) = (0.85, 0.118)$ (the convergence comparison of [NG13, Table 2]) and $(\alpha, r) = (0.95,\ 0.10/0.95) = (0.95, 0.105)$ (the ranking experiments of [NG13, §4]). We call the first the *canonical* setting because its total damping equals the canonical PageRank value; anchoring Stage 1 at $\alpha_0 = 0.85$ is our design choice, not that of [NG13].

2.3 Evaluation protocol in [NG13]

The ranking experiments of [NG13] use two perturbations of a complete Web graph. In the *newly added pages* test, a set of pages loses 90% of its incoming links. In the *sparseness* test, only 90%–40% of the links are kept. Each method's ranking on the perturbed graph is compared, with Kendall's τ, to *the same method's* ranking on the complete graph. This protocol measures stability. We show in Section 7.4 that, as a target for choosing parameters, it is degenerate.

For clarity, [NG13] Table 2 compares $(\eta, \mu) = (0.75, 0.10)$ with PageRank at 0.85 and $(0.85, 0.10)$ with PageRank at 0.95; its ranking experiments use total damping 0.95 against PageRank at 0.85. They do not test 0.85 versus 0.95 as a ranking-quality comparison, and the SELF protocol moves the reference with the candidate. The exact two-sided Wilcoxon p-value floor is 0.0625 for $n = 5$ and 0.0078 for $n = 8$.

2.4 Related work

Choosing the damping factor. Boldi, Santini and Vigna study PageRank as a function of α, give closed-form derivatives and show that $\alpha \to 1$ concentrates rank on irrelevant recurrent states [BSV05, BSV09]. Avrachenkov, Litvak and Pham analyse the ergodic structure of the Web graph with singular perturbation theory and recommend $\alpha \approx 1/2$ [ALP08]. Bressan and Peserico show that, in the worst case, the top-k ordering can take every permutation as α varies over an arbitrarily small interval [BP10]. Kamvar and Haveliwala show that the condition number of the PageRank problem grows like $(1 + \alpha)/(1 - \alpha)$ [KH03], and Haveliwala and Kamvar bound the second eigenvalue of the Google matrix by α [HK03]. Gleich et al. measure the teleportation parameter from browsing data [GCFG10]. Learned alternatives tune diffusion or teleportation coefficients from labelled data [BNG19]. We are not aware of prior work that sets α, or the NCDawareRank weights, from a closed-form structural statistic.

Damping-free rankings. TotalRank integrates PageRank over $\alpha \in [0, 1]$ [Bol05]. It is a special case of the generic damping functions of Baeza-Yates, Boldi and Castillo [BBC06], and it equals Random-α PageRank with $\alpha \sim \mathrm{Beta}(1, 1)$ [CG09]. In [NG13] these methods appear as competitors; here we use them as α-integrated references.

PageRank and in-degree. In networks without degree correlations, the mean PageRank of the pages with a given in-degree is an affine function of that in-degree [FBFM08], and PageRank and in-degree have power-law tails with the same exponent [LSV07]. The column sums of Z_r play the role of normalised in-degrees, and our statistic h measures how far they deviate from uniformity.

3 Stage 1: the block escape probability

3.1 Degeneracy of the escape probability under H

In earlier work [Kon26] the damping of a diffusion heuristic was calibrated from the mean probability that one step of the walk leaves the current vertex. Under H this statistic is degenerate: every non-dangling page sends all of its mass along its out-links, so the vertex-level escape probability is identically 1 (and $1 - 1/n$ for a dangling page) and carries no information about the graph. What does vary from page to page is *where* the mass goes, inside the home block or outside it, and under M this depends only on how many distinct blocks a page touches.

3.2 Escape probability under M

Let k_u be the number of distinct external blocks that u links to,

$$k_u = \left|\{\mathcal{A}_{(w)} : w \in \mathcal{G}_u,\ \mathcal{A}_{(w)} \neq \mathcal{A}_{(u)}\}\right|, \qquad \text{so that } N_u = k_u + 1.$$

Definition 1 (Block escape probability)**.** $P^{(M)}_{\text{esc}}(u) = 1 - \frac{1}{k_u+1} = \frac{k_u}{k_u+1}$, and $\bar{P}^{(M)}_{\text{esc}} = \frac{1}{n}\sum_{u\in\mathcal{W}} P^{(M)}_{\text{esc}}(u)$.

Proposition 1. *$P^{(M)}_{\text{esc}}(u)$ is the probability that one step of M from u leaves the home block $\mathcal{A}_{(u)}$.*

Proof. By (2), the row $M_{u\cdot}$ puts mass $1/N_u$ on each of the N_u proximal blocks, so the mass that stays in $\mathcal{A}_{(u)}$ is $1/N_u = 1/(k_u + 1)$. □

Remark 1 (Special partitions). With a single block, $M = \frac{1}{n}ee^\top$ and $v = u$ (the uniform vector), so NC-DawareRank reduces to PageRank with damping η, as noted in [NG13, Remark 1]. With n singleton blocks the same remark states that M coincides with H. Because u always belongs to $\mathcal{X}_u$, the exact statement is $M = (D + I)^{-1}(\text{Adj} + I)$ on non-dangling rows, the *lazy* hyperlink walk, and $M_{u\cdot} = e_u^\top$ on dangling rows. Our unit tests verify both forms.

3.3 The Stage 1 rule

Stage 1 fixes the total damping at $\alpha_0 = 0.85$ and sets the mixing ratio from the observed graph, $r = f(\bar{P}^{(M)}_{\text{esc}})$. The statistic summarises how strongly M couples the blocks: $\bar{P}^{(M)}_{\text{esc}} = 0$ when no page links outside its block, and $\bar{P}^{(M)}_{\text{esc}}$ grows towards 1 as pages touch many blocks. The functional form of f is not fixed in advance. We fit several candidate families on calibration graphs and lock the one that recovers the reference ranking best (Section 7). The locked rule is $f(\bar{P}^{(M)}_{\text{esc}}) = \text{clip}(0.242 - 0.143\,\bar{P}^{(M)}_{\text{esc}}, 0.01, 0.99)$. For the population truth POP we use the corresponding least-squares fit, $\text{clip}(0.420 - 0.386\,\bar{P}^{(M)}_{\text{esc}}, 0.01, 0.99)$.

4 Stage 2: the incoming-transition deviation

4.1 The ensembled incoming transition sum

For each page v, the incoming transition sums under the two logics are the column sums

$$\Sigma^{(H)}_{\text{in}}(v) = \sum_u H_{uv} = \sum_{u:\, v\in\mathcal{G}_u} \frac{1}{d_u} + \frac{n_{\text{d}}}{n}, \tag{4}$$

$$\Sigma^{(M)}_{\text{in}}(v) = \sum_u M_{uv} = \frac{1}{|\mathcal{A}_{(v)}|} \sum_{u:\, \mathcal{A}_{(v)}\subseteq\mathcal{X}_u} \frac{1}{N_u}, \tag{5}$$

where n_{d} is the number of dangling pages, each of which contributes $1/n$ to every column. $\Sigma^{(M)}_{\text{in}}$ is constant within a block, and it equals the row sum of $R^\top$ at that block. The *ensembled* incoming sum is the column sum of Z_r,

$$\Sigma_{\text{in}}(v; r) = r\,\Sigma^{(M)}_{\text{in}}(v) + (1 - r)\,\Sigma^{(H)}_{\text{in}}(v). \tag{6}$$

Proposition 2 (Mean incoming sum)**.** $\frac{1}{n}\sum_v \Sigma_{\text{in}}(v;r) = 1$ *for every* r.

Proof. $\sum_v \Sigma_{\text{in}}(v;r) = \sum_u \sum_v (Z_r)_{uv} = n$, because Z_r is row-stochastic. □

The identity holds only when the dangling correction in (4) is included; without it the mean is $1 - n_{\text{d}}/n$. Across all instances and r-settings of our experiments, the largest deviation of the mean from 1 is 3.3×10^{-16}.

4.2 The meaning of the deviation statistic

Definition 2 (Deviation statistics)**.**

$$h(r) = \frac{1}{n}\sum_{v\in\mathcal{W}} \left|1 - \Sigma_{\text{in}}(v;r)\right|, \qquad h_v(r) = \left\|v^\top Z_r - v^\top\right\|_1.$$

Proposition 3 (Interpretation)**.** *Let* $u = \frac{1}{n}e$ *be the uniform distribution. Then*

(i) $h(r) = \|u^\top Z_r - u^\top\|_1$*: h is the L_1 distance that the uniform distribution moves in one step of Z_r, and $h = 0$ if and only if Z_r is doubly stochastic;*

(ii) $h_v(r) = \left\|\frac{\mathrm{d}}{\mathrm{d}\alpha}\pi(\alpha)\big|_{\alpha=0}\right\|_1$.

Proof. (i) $(u^\top Z_r)_v = \frac{1}{n}\sum_w (Z_r)_{wv} = \Sigma_{\text{in}}(v;r)/n$, so $\|u^\top Z_r - u^\top\|_1 = \frac{1}{n}\sum_v |\Sigma_{\text{in}}(v;r) - 1|$, which vanishes if and only if every column sum equals 1. (ii) From $\pi(\alpha)^\top = (1-\alpha)\sum_k \alpha^k v^\top Z_r^k = v^\top + \alpha(v^\top Z_r - v^\top) + O(\alpha^2)$, the derivative at 0 is $v^\top Z_r - v^\top$. The same power series of PageRank in α is the starting point of the analysis in [BSV09]. □

Proposition 4 (When the ranking equals the prior)**.** *For any $\alpha \in (0,1)$: $\pi(\alpha) = v$ if and only if $h_v(r) = 0$. If all blocks have the same size, then $v = u$, $h_v = h$, and $\pi(\alpha) = u$ if and only if $h = 0$. If block sizes differ, $h = 0$ (that is, Z_r doubly stochastic) does* not *imply $\pi(\alpha) = u$. In that case $\|\pi(\alpha) - u\|_1 \le \|v - u\|_1$.*

Proof. If $v^\top Z_r = v^\top$, every term of the series equals $v^\top$, so $\pi = v$. Conversely, the fixed-point equation $\pi^\top = \alpha\pi^\top Z_r + (1-\alpha)v^\top$ with $\pi = v$ gives $\alpha v^\top = \alpha v^\top Z_r$. If $u^\top Z_r = u^\top$, then $\pi(\alpha)^\top - u^\top = (1-\alpha)(v-u)^\top(I - \alpha Z_r)^{-1}$. The matrix $(1-\alpha)(I-\alpha Z_r)^{-1} = (1-\alpha)\sum_k \alpha^k Z_r^k$ is row-stochastic, and a row-stochastic matrix does not increase the L_1 norm of a row vector (Lemma 1). □

Thus h describes the chain under uniform teleportation, whereas under the block-even teleportation of NC-DawareRank the relevant quantity is h_v.

4.3 A displacement bound

Lemma 1. *For every row vector x and every row-stochastic S, $\|x^\top S\|_1 \le \|x\|_1$.*

Proof. $\|x^\top S\|_1 = \sum_j \left|\sum_i x_i S_{ij}\right| \le \sum_i |x_i| \sum_j S_{ij} = \|x\|_1$. □

Theorem 1 (Displacement bound)**.** *For every $r \in [0,1]$ and $\alpha \in [0,1)$,*

$$\|\pi(\alpha) - v\|_1 \;\le\; \frac{\alpha}{1-\alpha}\, h_v(r), \tag{7}$$

and $\|\pi(\alpha) - v\|_1 = \alpha\, h_v(r) + O(\alpha^2)$ as $\alpha \to 0$, so the bound is tight to first order.

Proof. Since $\sum_{k\ge 0}(1-\alpha)\alpha^k = 1$, $\pi(\alpha)^\top - v^\top = \sum_{k\ge 0}(1-\alpha)\alpha^k\,(v^\top Z_r^k - v^\top)$. Each bracket telescopes, $v^\top Z_r^k - v^\top = \sum_{j=0}^{k-1}(v^\top Z_r - v^\top)Z_r^j$, and by Lemma 1 every summand has L_1 norm at most $\|v^\top Z_r - v^\top\|_1 = h_v(r)$. Hence $\|v^\top Z_r^k - v^\top\|_1 \le k\, h_v(r)$ and

$$\|\pi(\alpha) - v\|_1 \le h_v(r)\sum_{k\ge 0}(1-\alpha)\alpha^k k = h_v(r)\,\frac{\alpha}{1-\alpha}.$$

The first-order statement follows from Proposition 3(ii). □

Remark 2 (What the bound does and does not say). Because $\|\pi - v\|_1 \le 2$, the useful form is $\|\pi(\alpha) - v\|_1 \le \min\{2, \alpha h_v/(1-\alpha)\}$. The bound limits how far the ranking can move away from the teleportation prior, and h_v is the rate at which it starts to move. It says nothing about which α recovers a given ranking best. Numerically, on 14 graphs (2 948 combinations of graph, r and α), the ratio of the left to the right side of (7) never exceeds 0.961. It lies between 0.94 and 0.96 at $\alpha = 0.05$, and its median is 0.16 at $\alpha = 0.85$: the bound is tight for small α and conservative by a factor of about six at the usual damping.

4.4 Calibrating $\alpha = g(h)$ and recovering (η, μ)

Stage 2 keeps the ratio r of Stage 1, computes $h(r)$ (or $h_v(r)$) on the observed graph and sets $\alpha = g(h)$, after which $\eta = (1 - r)\alpha$ and $\mu = r\alpha$. The working hypothesis behind this stage was that a larger deviation from double stochasticity calls for a larger α. We test this hypothesis, its opposite and several functional forms for g, including the one-parameter *displacement-budget* rule $\alpha = c/(c + h_v)$ suggested by Theorem 1: under it the right-hand side of (7) equals the budget c for every graph (Section 8).

5 The complete procedure and its cost

Algorithm 1 states the procedure. Both statistics are by-products of building R, and the procedure adds no asymptotic cost to NCDawareRank.

Proposition 5 (Cost)**.** *Given the arc list and the block map, $\{k_u\}$, $\bar{P}^{(M)}_{\text{esc}}$, $\Sigma^{(H)}_{\text{in}}$, $\Sigma^{(M)}_{\text{in}}$, $h(r)$ and $h_v(r)$ are computable in $O(|\mathcal{E}| + n)$ time with $O(n + N)$ memory beyond the graph.*

Proof. Keep an array last$[1..N]$, initially zero. For each page u, visit its own block and then the block of every out-neighbour; a block I is new for u exactly when last$[I] \neq u$, in which case we set last$[I] \leftarrow u$ and record the pair (u, I). Page u costs $O(1 + d_u)$, so all non-zeros of R, which must be enumerated to build R anyway, are found in $O(|\mathcal{E}| + n)$ time, and $\text{nnz}(R) \le n + |\mathcal{E}|$. Counting them per page gives N_u, hence k_u and $\bar{P}^{(M)}_{\text{esc}}$, in $O(n)$. One pass over the arcs accumulates $1/d_u$ into $\Sigma^{(H)}_{\text{in}}$, and the dangling term adds n_{d}/n to every entry in $O(n)$. $\Sigma^{(M)}_{\text{in}}$ accumulates $1/N_u$ into each proximal block in $O(\text{nnz}(R))$ and is divided by the block sizes and copied to the pages in $O(n)$. $h(r)$ is one $O(n)$ pass. $h_v(r)$ needs the product $v^\top Z_r = (1 - r)\, v^\top H + r\,(v^\top R)A$, which costs $O(\text{nnz}(H) + \text{nnz}(R) + n)$, the same as one power iteration. The extra memory is the array last and the block sums, $O(N)$, and a constant number of page vectors, $O(n)$. □

The functional forms of f and g are fitted on calibration graphs (Sections 7 and 8). The rules locked by those experiments are

$$r = f(\bar{P}^{(M)}_{\text{esc}}) = \text{clip}(0.242 - 0.143\, \bar{P}^{(M)}_{\text{esc}},\ 0.01,\ 0.99), \qquad \alpha = g(h) \equiv 0.810. \tag{8}$$

The constant form of g is not a simplification we imposed. It is the rule that maximises rank agreement among all fitted forms, as Section 8 shows. When the reference ranking is the setting of the original experiments, $(\alpha, r) = (0.95, 0.105)$, the corresponding constant is $\alpha = 0.930$.

6 Experimental methodology

6.1 Synthetic block webs

Static fitness generator. Pages are partitioned into blocks with log-normal sizes (mean block size log-uniform in $[4, 100]$), and blocks are grouped into domains. Each page has a latent quality q_u with $\log q_u = s_b z_{\mathcal{A}_{(u)}} + s_w z_u$. The total dispersion σ_q lies in $[0.8, 2]$, and the share of block-level variance (the intraclass correlation, ICC) in $[0, 0.8]$. Out-degrees are log-normal (mean in $[2, 25]$), and a fraction 0.02–0.30 of pages is dangling. An arc stays in the source's block with probability $\varphi \in [0.30, 0.97]$ and then targets a page in proportion to q. Otherwise it goes to another block, chosen in proportion to a block attractiveness Q_B^β times a log-normal factor, either inside the domain (probability $\lambda \in [0, 0.9]$) or globally, and then to a page of that block in proportion to q. Page identifiers are shuffled. The generator's expected walk is known in closed form, and its PageRank defines the population truth POP (Section 6.5).

Algorithm 1 Two-stage calibrated NCDawareRank

Require: arcs $\mathcal{E}$, block map $u \mapsto \mathcal{A}_{(u)}$, rules f, g, tolerance ε

build H (dangling rows uniform), the non-zeros of R, A and v ▷ $O(|\mathcal{E}| + n)$

$k_u \leftarrow N_u - 1$; $\bar{P}^{(M)}_{\text{esc}} \leftarrow \frac{1}{n}\sum_u k_u/(k_u+1)$; $r \leftarrow f(\bar{P}^{(M)}_{\text{esc}})$ ▷ Stage 1

$\Sigma^{(H)}_{\text{in}} \leftarrow$ column sums of H; $\Sigma^{(M)}_{\text{in}}(v) \leftarrow (R^\top e)_{\mathcal{A}_{(v)}}$

$h \leftarrow \frac{1}{n}\sum_v |1 - r\Sigma^{(M)}_{\text{in}}(v) - (1-r)\Sigma^{(H)}_{\text{in}}(v)|$; $\alpha \leftarrow g(h)$ ▷ Stage 2

$\eta \leftarrow (1-r)\alpha$; $\mu \leftarrow r\alpha$; $\pi \leftarrow v$

repeat

  $\pi' \leftarrow \eta\,\pi^\top H + \mu\,(\pi^\top R)A + (1-\alpha)v^\top$ ▷ $O(\text{nnz}(H) + \text{nnz}(R) + n)$

  $\delta \leftarrow \|\pi' - \pi\|_1$; $\pi \leftarrow \pi'$

until $\delta < \varepsilon$

return π

Table 1: Real networks used for transfer. Self-loops and duplicate arcs removed. $\bar{P}^{(M)}_{\text{esc}}$ is computed on the complete graph.

graph	NCD blocks	n	arcs	blocks	intra-block arcs (%)	dangling (%)	$\bar{P}^{(M)}_{\text{esc}}$
cnr-2000	host proxy (contiguous URL-order segments)	325 557	3 128 710	7 604	89.4	26.7	0.110
WikiCS	10 topic labels	11 701	290 519	10	68.8	3.8	0.506
Cora (full)	70 topic labels	19 793	65 311	70	57.1	11.0	0.335
Cora-ML	7 topic labels	2 995	8 416	7	79.2	11.7	0.166
CiteSeer	6 topic labels	4 230	5 358	6	94.9	21.3	0.026

Growth generator. Pages arrive one at a time. Each either founds a new site or joins an existing one with probability proportional to its size, and links to earlier pages by preferential attachment on in-degree, inside its own site with probability φ. A final update phase adds 10%–50% more arcs from existing pages. The mechanism differs from the static generator (age effects, Yule-type site sizes), and we use it only for transfer.

6.2 Real networks

Table 1 lists the five networks. cnr-2000 is the Italian CNR crawl of the LAW collection [BV04, BCSV04] and one of the two graphs used in the ranking experiments of [NG13]. We decoded it from its BVGraph files. Its URL list was not available, so the host partition of [NG13] cannot be reproduced exactly. Because LAW numbers pages in lexicographic URL order, every host is an interval of identifiers, and we recover a *host proxy* by greedy, contiguity-constrained maximisation of directed modularity. It yields 7 604 segments, with 89% of the arcs inside segments. WikiCS [MC20] and the CitationFull graphs Cora, Cora-ML and CiteSeer [BG18] are directed and carry topic labels, which we use as blocks. Content category is one of the alternative decomposition criteria suggested in [NG13, §5]. The real graphs span $\bar{P}^{(M)}_{\text{esc}} \in [0.03, 0.51]$ on the complete graph, with cnr-2000 at 0.110.

6.3 Perturbations

Following [NG13, §4.2], each base graph yields perturbed instances. In *Experiment A* (sparsity), a uniformly random fraction $p \in \{0.9, 0.8, \ldots, 0.2\}$ of the arcs is kept. In *Experiment B* (newly added pages), a fraction $f \in \{0.1, \ldots, 0.6\}$ of the pages is chosen and each of their incoming arcs is deleted with probability 0.9. Every perturbation is followed by rebuilding H, R and A on the observed graph. Calibration graphs and the four smaller real graphs have two replicates per level, and the other panels one. The unperturbed graph (A1.0) is also kept.

Table 2: Experimental panels. Every instance is one perturbation of one base graph (Experiments A and B, Section 6.3); splits are by base graph. The instance counts are those of the Stage 2 replay and include the unperturbed graph A1.0.

panel	generator	role	graphs	instances	n
`calib`	static fitness block web	fitting	64	1 856	2 000
`test`	static (independent seeds)	held-out	32	480	2 000
`scale`	static, 8× larger	scale transfer	12	180	16 000
`topo`	growth / preferential attachment	topology transfer	24	360	3 000
`real`	Table 1	real-network transfer	5	131	2 995–325 557

6.4 Panels and splits

Table 2 summarises the panels. All splits are by base graph, and no base graph contributes to more than one panel. Rules are fitted on `calib` only. `test` uses the same generator with independent seeds, `scale` uses graphs eight times larger, `topo` uses the growth generator, and `real` uses the five networks.

6.5 Ground truths

A ranking method is scored by Kendall's τ between its ranking of the *observed* (perturbed) graph and a reference ranking. Each reference is computed on the *complete* base graph, except Q, which is latent.

NCD canonical NCDawareRank, $(\alpha, r) = (0.85, 0.118)$; the primary truth.

NCD95 the setting of the experiments of [NG13], $(\alpha, r) = (0.95, 0.105)$.

PR PageRank ($r = 0$) at $\alpha = 0.85$ with block-even teleportation.

SELF the protocol of [NG13]: the candidate's own ranking on the complete graph, with the same (α, r).

POP PageRank at 0.85 of the static generator's expected walk: a population importance that no ranking method defines.

Q the latent quality q of the static generator, which is free of any walk.

TOT the α-integrated NCDawareRank with $r = 0.118$, $\int_0^1 \pi(\alpha)\,\mathrm{d}\alpha = \sum_k v^\top Z_r^k/((k+1)(k+2))$, that is, TotalRank [Bol05] or Random-α NCDawareRank with a Beta(1, 1) prior [CG09]. Its implicit prior has mean 0.5, so it is an α-integrated reference rather than a neutral one.

A Random-α family $\mathbb{E}[\pi(A)]$, $A \sim \mathrm{Beta}(a, b)$, with weights $w_k = B(a + k, b + 1)/B(a, b)$, is used for the dose-response test of Section 8.3.

6.6 Oracle parameters and solvers

For each instance, the oracle ratio r^* maximises τ over $r \in \{0.01, \ldots, 0.99\}$ at $\alpha_0 = 0.85$. The oracle damping α^* maximises τ over $\alpha \in \{0.40, 0.41, \ldots, 0.99\}$ with r locked; seven coarser points in $[0.05, 0.35]$ are computed for diagnostics. References and Stage 1 rankings use the factorised power iteration with L_1 tolerance 10^{-10}–10^{-12}. For Stage 2, all 67 values of α come from a single Neumann series per r,

$$\pi(\alpha_j)^\top \approx (1 - \alpha_j) \sum_{k=0}^{K_j} \alpha_j^k\, u_k + \alpha_j^{K_j+1} u_{K_j+1}, \qquad u_k = v^\top Z_r^k,$$

with $K_j = \lceil \ln(10^{-13}) / \ln \alpha_j \rceil$. The truncation error is at most $2 \cdot 10^{-13}$ in L_1. The series agrees with the power iteration to 10^{-11} and reproduces the Stage 1 values of τ at $\alpha = 0.85$ to 10^{-6}. Both methods use only matrix–vector products and entry-independent scalings, so pages with identical in-neighbourhoods keep bit-identical scores. This matters for Kendall's τ: a pivoted direct solver breaks such structural ties with rounding noise, and on our $n = 2\,000$ graphs this lowered τ by 0.002–0.004, a sizeable fraction of the effects studied here.

Table 3: Stage 1 candidate rules $r = f(\bar{P})$, $\bar{P} = \bar{P}_{\text{esc}}^{(M)}$, at $\alpha_0 = 0.85$, truth NCD. Columns `test–real`: mean Kendall τ at the rule's r. CV τ: grouped 8-fold cross-validation over calibration graphs. Gap: share of the canonical-to-oracle gap closed on `test`. p: Wilcoxon signed-rank test of the per-graph mean differences against the constant rule on `test`. Clip-linear rules are clipped to $[0.01, 0.99]$; LS: least squares on the oracle values; τ-opt.: fitted by maximising the mean calibration τ. The logistic and isotonic rules are reported for completeness.

rule	formula	CV τ	`test`	`scale`	`topo`	`real`	gap %	p
constant r	0.180	0.6854	0.6864	0.6681	0.7838	0.7586	33	–
clip-linear, LS	clip(0.304 − 0.156 $\bar{P}$)	0.6831	0.6855	0.6656	0.7816	0.7523	28	0.443
clip-linear, τ-opt. (locked)	clip(0.242 − 0.143 $\bar{P}$)	0.6859	0.6867	0.6681	0.7844	0.7601	34	0.331
logistic, τ-opt.	–	0.6860	0.6868	0.6680	0.7845	0.7599	34	0.360
isotonic	–	0.6832	0.6859	0.6650	0.7818	0.7522	30	0.612
canonical $r = 0.118$			0.6801	0.6633	0.7840	0.7559	0	
oracle r^*			0.6995	0.6788	0.7905	0.7685	100	

6.7 Statistics and candidate rules

Kendall's τ-b [Ken38, Ken45] is the primary metric, the one used in [NG13], and Spearman's ρ is secondary. The optimal parameters chosen by the two metrics agree closely (Pearson correlation 0.960 for r^*). Perturbed instances of one base graph are not independent, so every paired comparison uses per-graph means. We report Wilcoxon signed-rank p-values [Wil45] and bootstrap percentile intervals over graphs (20 000 resamples [ET93]). Model selection uses 8-fold cross-validation grouped by graph. The candidate families for f and g are:

- a constant;
- clipped-linear rules $\text{clip}(c_0 + c_1 x, \ell, u)$, fitted by least squares on the oracle values or directly by maximising the mean calibration τ over the stored τ curves ("τ-optimised");
- the same with free bounds;
- logistic rules;
- isotonic regression;
- for Stage 2, the displacement-budget rule $c/(c + x)$;
- as diagnostic ceilings, a two-feature linear rule and gradient-boosted trees [Fri01] on every observable global statistic of the observed graph.

Decision-focused (τ-optimised) fitting is robust to flat optima and to oracle values that sit at the grid bounds.

7 Results: Stage 1, the mixing ratio

7.1 Candidate rules

Figure 1 plots the oracle ratio r^* against $\bar{P}_{\text{esc}}^{(M)}$ for the 64 calibration graphs, and Table 3 reports every candidate. Under the canonical truth, the τ-optimised clipped-linear rule is $r = \text{clip}(0.242 - 0.143\,\bar{P}_{\text{esc}}^{(M)}, 0.01, 0.99)$. Its cross-validated τ (0.6859) ties with the logistic rule (0.6860) and exceeds the constant rule $r = 0.180$ (0.6854) and the least-squares rule (0.6831). On the held-out `test` panel the rule reaches $\tau = 0.6867$ against 0.6864 for the constant, closing 34% of the canonical-to-oracle gap (τ from 0.6801 to 0.6995). The difference from the constant is not significant ($p = 0.331$). The constant itself beats the canonical r by +0.0063 (95% CI $[0.0041, 0.0085]$, 30/32 graphs, $p = 6.0 \times 10^{-6}$). On `topo`, the rule is significantly better than the constant (+0.0007, $p = 0.011$, which survives a Holm correction over the four transfer panels), but the effect is small. There the constant does not differ significantly from the canonical setting (−0.0002, $p = 0.065$). On the real networks, the rule (0.7601) and the constant (0.7586) are statistically indistinguishable ($p = 0.812$).

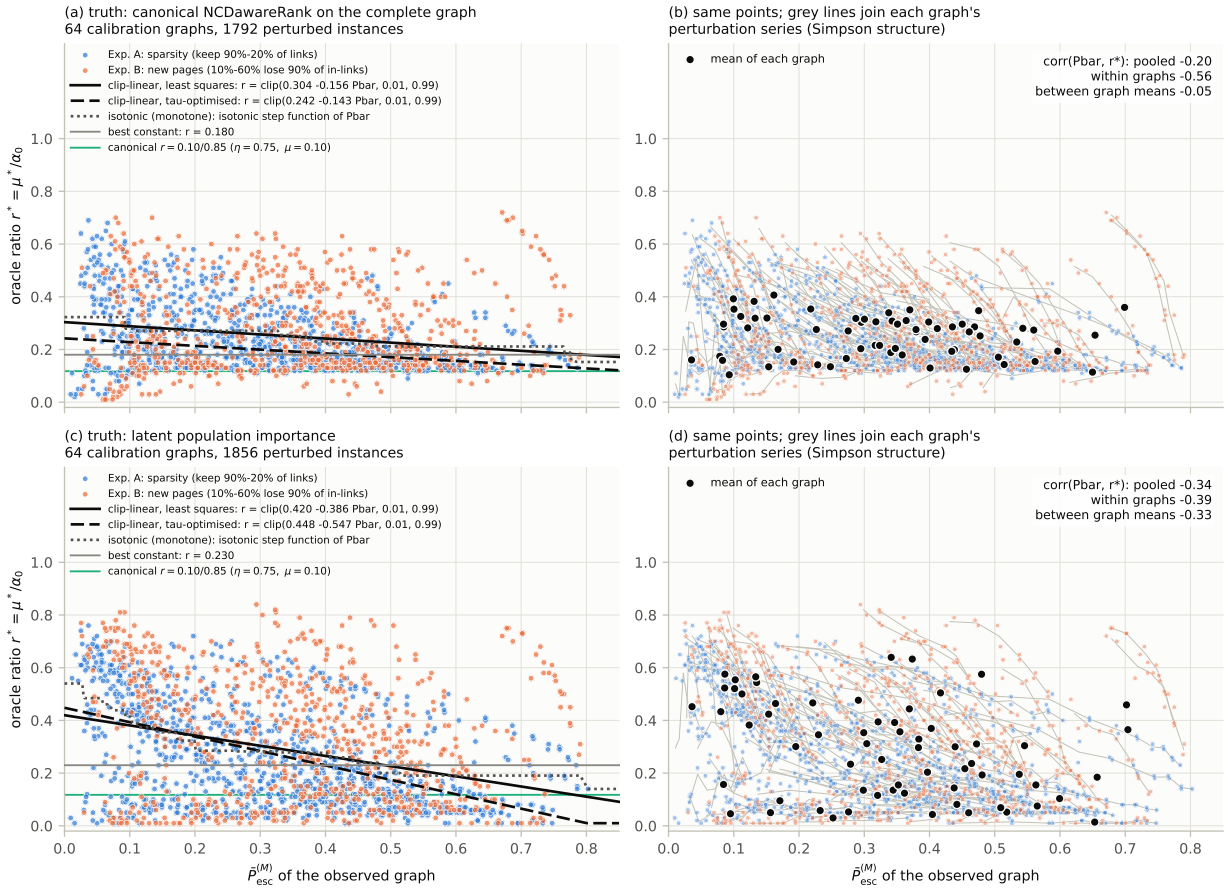


Figure 1: Stage 1: oracle ratio r^* against $\bar{P}^{(M)}_{\text{esc}}$ of the observed graph on the calibration panel. **(a)** Truth NCD with the fitted candidate rules and the canonical r. **(c)** Truth POP. **(b, d)** The same points, with grey lines joining the perturbation series of each graph and black dots marking graph means. The pooled slope is a mixture of a steep within-graph relation and an almost flat between-graph one.

7.2 Why $\bar{P}^{(M)}_{\text{esc}}$ does not transfer: Simpson structure

The pooled correlation between $\bar{P}^{(M)}_{\text{esc}}$ and r^* is -0.204. Within graphs it is -0.560 (-0.763 in Experiment A and -0.769 in Experiment B), and between graph means it is -0.054. Inside a graph, $\bar{P}^{(M)}_{\text{esc}}$ simply tracks how many links the perturbation removed: its within-graph correlation with the fraction of arcs kept is $+0.936$ (A) and $+0.956$ (B). Removing links lowers both $\bar{P}^{(M)}_{\text{esc}}$ and the agreement between observed and complete graph, and the latter calls for more weight on the block-level smoothing of M. Across graphs, however, the level of $\bar{P}^{(M)}_{\text{esc}}$ says almost nothing about r^*. A practitioner observes a single graph and cannot tell whether a low $\bar{P}^{(M)}_{\text{esc}}$ comes from heavy link loss or from a naturally block-confined graph. The fitted slope is therefore of little use out of sample.

7.3 What determines r^*

Table 4 decomposes the variance of r^* under each truth. Under NCD, graph identity explains 0.31 and the perturbation level 0.42; additively they explain 0.74. Boosted trees on the *hidden* generator parameters plus the level recover 0.71 out of sample, whereas every observable statistic together recovers -0.01 and $\bar{P}^{(M)}_{\text{esc}}$ alone 0.03. The graph-level component is associated with the latent quality structure. The graph mean of r^* correlates $+0.548$ with the ICC and -0.578 with the dispersion of latent quality under NCD, and $+0.527$ and -0.597 under POP (Figure 3c). This is the regime in which shrinkage towards block averages, which is what M performs, pays off. Neither quantity is visible in global statistics of the observed graph. The test–retest correlation of r^* between replicates is 0.920, so the unexplained part is not noise in the oracle.

Table 4: Share of the variance of the oracle ratio r^* explained (R^2) on the calibration panel. Graph identity, perturbation level and their additive combination are in-sample ANOVA shares. Starred columns are out of sample (grouped 8-fold CV by graph): boosted trees on the hidden generator parameters plus the perturbation level; ridge regression on every observable global statistic; and the clip-linear rule on $\bar{P}_{\text{esc}}^{(M)}$. Negative values mean worse than predicting the mean.

truth	graph	level	graph+level	hidden+level*	all observables*	$\bar{P}_{\text{esc}}^{(M)}$ alone*
NCD	0.31	0.42	0.74	0.71	-0.01	0.03
PR	0.41	0.30	0.71	0.70	-0.08	-0.02
POP	0.74	0.13	0.87	0.63	0.11	0.07
SELF	0.34	0.04	0.38	-0.36	-0.02	0.00

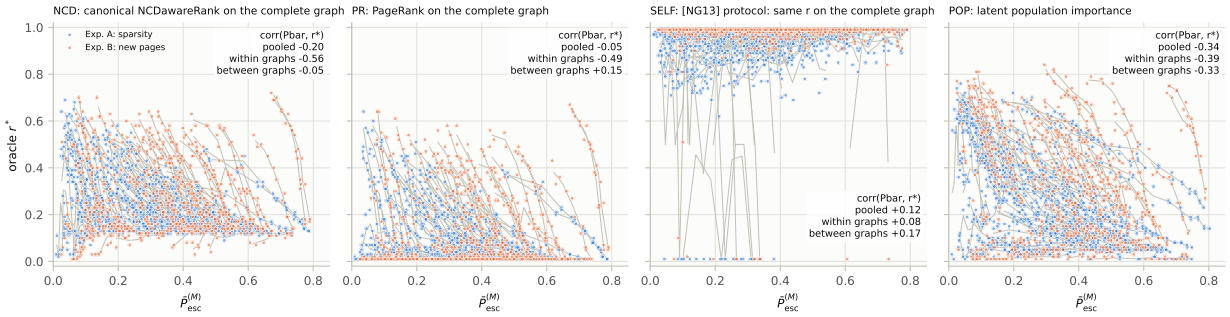


Figure 2: Stage 1 oracle ratio against $\bar{P}_{\text{esc}}^{(M)}$ under the four truths of Stage 1 (calibration panel). Under SELF, the protocol of [NG13], the oracle is pushed to the upper bound of the grid.

7.4 Sensitivity to the ground truth

Figure 2 repeats the analysis under all four truths. Under PR, 38% of the oracle ratios sit at the lower bound $r = 0.01$, because a PageRank reference favours $r \to 0$. Under POP, the relation with $\bar{P}_{\text{esc}}^{(M)}$ is negative both within graphs (-0.387) and between graphs (-0.334), but the fitted rule still does not beat a constant out of sample (Appendix Table 17). Under SELF, the protocol of [NG13], 51% of the oracle ratios sit at the upper bound $r = 0.99$. Comparing a ranking with *its own* complete-graph ranking rewards whatever makes the ranking insensitive to links: large r smooths rank over blocks, which are not perturbed. The same holds for α: as $\alpha \to 0$, both rankings tend to v, which depends only on the partition, so $\tau \to 1$. Stability is a legitimate property to report, but it cannot serve as a target for choosing parameters.

7.5 Real networks

Per graph, the Stage 1 rule changes τ relative to the canonical setting by +0.0325 (CiteSeer), -0.0071 (cnr-2000), -0.0053 (Cora), -0.0055 (Cora-ML) and +0.0006 (WikiCS). The panel-level gain comes from CiteSeer alone, a very sparse graph (mean out-degree 1.27) whose arcs fall 95% inside blocks. The fitted rule is also extrapolated there: CiteSeer's $\bar{P}_{\text{esc}}^{(M)} = 0.026$ lies at the edge of the calibration range. On cnr-2000 ($\bar{P}_{\text{esc}}^{(M)} = 0.110$) the rule prescribes $r = 0.226$, i.e. $\mu = 0.192$ at $\alpha = 0.85$, about twice the canonical μ, and loses τ against it. The rule returns $r \approx 0.12$ only for $\bar{P}_{\text{esc}}^{(M)} \approx 0.85$, a value no real graph in this study approaches.

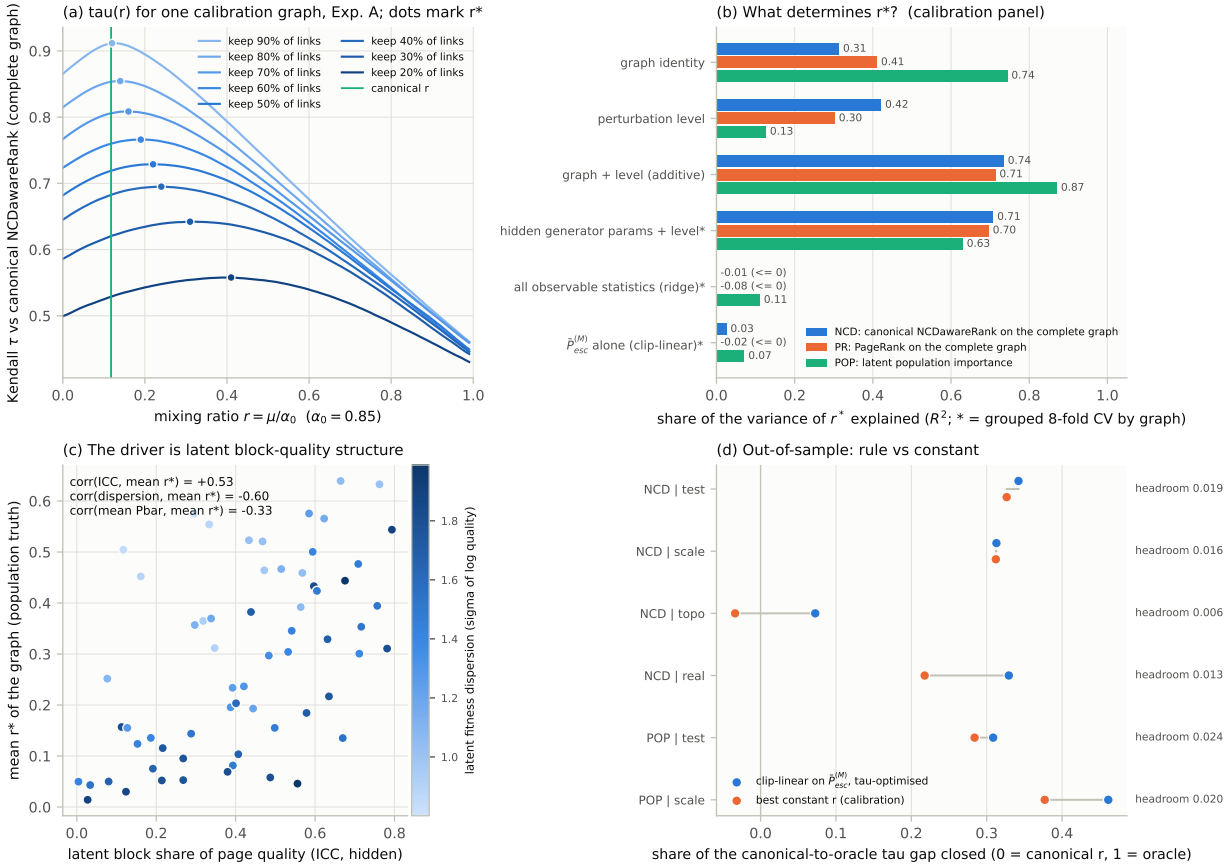


Figure 3: Stage 1 diagnostics. **(a)** $\tau(r)$ curves for one calibration graph under increasing link loss; dots mark r^*, which moves away from the canonical value as links disappear. **(b)** Share of the variance of r^* explained by each source. **(c)** Graph-mean r^* under POP against the latent ICC, coloured by the dispersion of latent quality. **(d)** Share of the canonical-to-oracle gap closed out of sample by the rule and by the best constant.

8 Results: Stage 2, the total damping

8.1 Candidate rules

Figure 4(a) plots the oracle damping α^* against h for the calibration graphs, with r locked to the Stage 1 rule; Table 5 reports every candidate. The oracle is $\alpha^* = 0.791 \pm 0.058$. The τ-optimised clipped-linear rule converges to the constant $\alpha = \text{clip}(0.810 - 0.000\,h) = 0.810$: once rank agreement is the objective, the slope on h is driven to zero. Every candidate that does depend on h is worse than the constant.

- The least-squares rule $\text{clip}(0.803 - 0.021\,h)$ loses -0.0015 on `test` ($p = 1.8 \times 10^{-4}$) and -0.0007 on `topo` ($p = 7.4 \times 10^{-4}$).
- The isotonic rule loses -0.0016 ($p = 0.002$) on `test`.
- The displacement-budget rule $c/(c + h)$, $c = 2.69$, loses -0.0023 on `test` and -0.0043 on `scale`.
- Boosted trees on every observable statistic, including h and h_v, are the most flexible candidate, yet they lose -0.0018 on `topo` ($p = 8.5 \times 10^{-4}$).
- Replacing h by h_v changes nothing (the least-squares rule $\text{clip}(0.802 - 0.016\,h_v)$ loses -0.0015 on `test`, $p = 2.1 \times 10^{-4}$).

The constant itself improves on $\alpha = 0.85$ by $+0.0044$ on `test` ($p = 1.3 \times 10^{-6}$), $+0.0054$ on `scale` and $+0.0031$ on `topo`, closing 47%, 51% and 46% of the gap to the oracle. On `real` it changes τ by -0.0012 ($p = 1.000$).

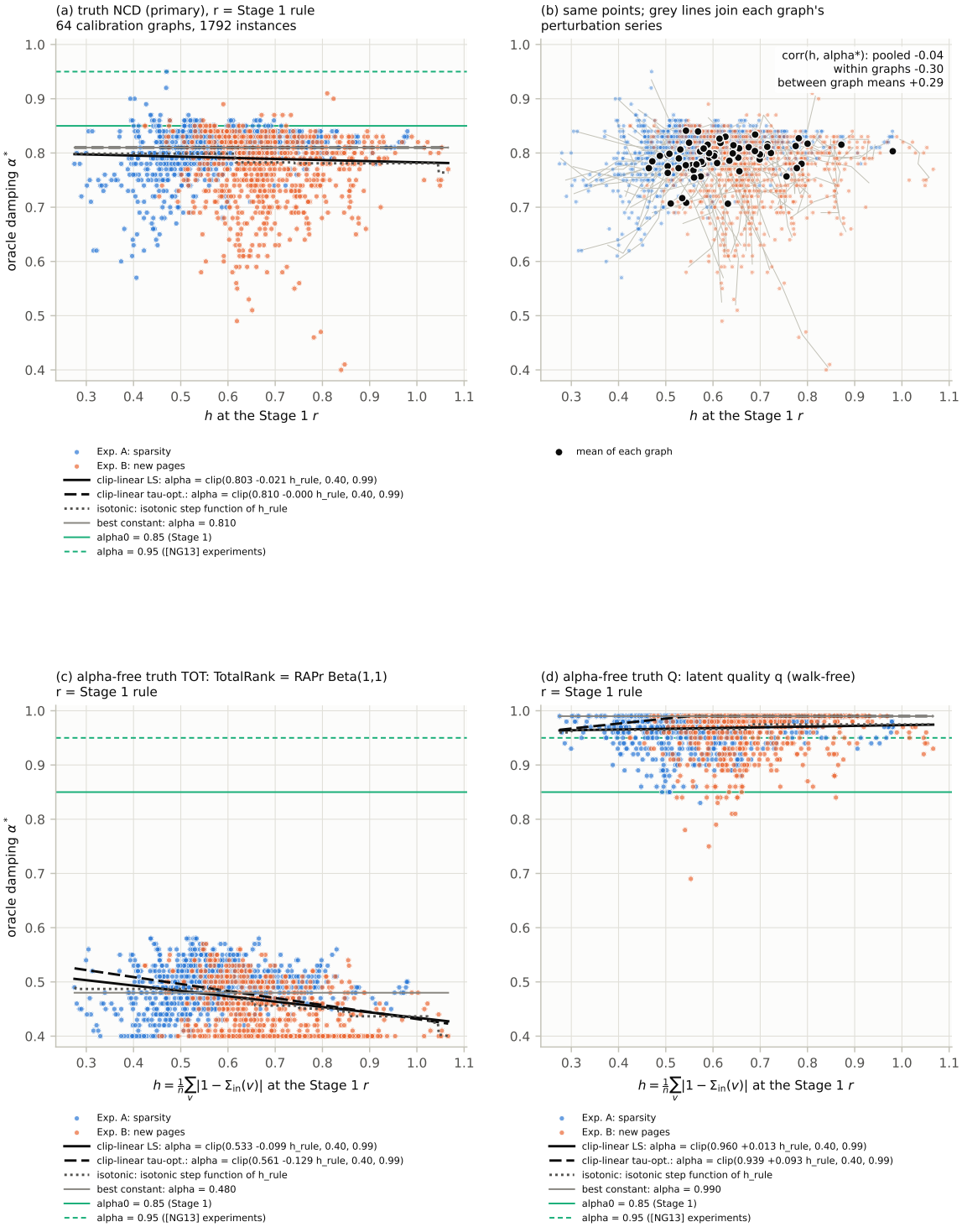


Figure 4: Stage 2: oracle damping α^* against h (computed at the Stage 1 r) on the calibration panel. **(a)** Primary truth NCD with the fitted candidates of g. The least-squares and τ-optimised fits are nearly flat, and the latter coincides with the constant 0.810. **(b)** The same points with each graph's perturbation series joined. **(c, d)** The two α-free truths, TOT ($\alpha^* \approx 0.47$) and Q (α^* near the top of the grid). Horizontal lines: $\alpha_0 = 0.85$ and the $\alpha = 0.95$ of [NG13].

Table 5: Stage 2 candidate rules $\alpha = g(h)$ and $\alpha = g(h_v)$ with r locked to the Stage 1 rule, truth NCD. Top block: mean Kendall τ of the static damping, the locked constant, their difference and the oracle. Lower blocks: difference in mean τ of each fitted rule against the locked constant. CV τ: grouped 8-fold cross-validation on the calibration graphs. †: Wilcoxon signed-rank $p < 0.01$ over graphs. Every rule is clipped to $[0.40, 0.99]$; LS: least squares on the oracle values; τ-opt.: fitted by maximising the mean calibration τ.

rule	formula	CV τ	test	scale	topo	real
mean Kendall τ						
static $\alpha = 0.85$		–	0.6867	0.6681	0.7844	0.7600
constant $\alpha = 0.810$ (locked)		0.6913	0.6911	0.6735	0.7875	0.7559
difference			$+0.0044^{\dagger}$	$+0.0054^{\dagger}$	$+0.0031^{\dagger}$	-0.0012
oracle α^*		–	0.6960	0.6787	0.7911	0.7738
rules on h: difference in mean τ against the locked constant						
clip-linear, τ-opt. (locked)	$\mathrm{clip}(0.810 - 0.000\,h)$	0.6912	0.0000	0.0000	0.0000	0.0000
clip-linear, LS	$\mathrm{clip}(0.803 - 0.021\,h)$	0.6898	$-0.0015^{\dagger}$	-0.0013	$-0.0007^{\dagger}$	-0.0033
isotonic	–	0.6897	$-0.0016^{\dagger}$	-0.0014	$-0.0010^{\dagger}$	-0.0049
budget $c/(c + x)$, τ-opt.	$c = 2.69$	0.6891	$-0.0023^{\dagger}$	$-0.0043^{\dagger}$	$-0.0016^{\dagger}$	-0.0045
boosted trees, all observables	–	0.6905	-0.0002	-0.0030	$-0.0018^{\dagger}$	-0.0124
rules on h_v: difference in mean τ against the locked constant						
clip-linear, τ-opt. (locked)	$\mathrm{clip}(0.810 + 0.000\,h_v)$	0.6912	0.0000	0.0000	0.0000	0.0000
clip-linear, LS	$\mathrm{clip}(0.802 - 0.016\,h_v)$	0.6898	$-0.0015^{\dagger}$	-0.0013	$-0.0011^{\dagger}$	-0.0029
isotonic	–	0.6896	$-0.0018^{\dagger}$	-0.0013	$-0.0024^{\dagger}$	-0.0037
budget $c/(c + x)$, τ-opt.	$c = 3.04$	0.6888	-0.0024	-0.0037	$-0.0083^{\dagger}$	-0.0019

8.2 Why h cannot track α^*

The pooled correlation between h and α^* is -0.045. Within graphs it is -0.304, and between graph means $+0.291$ (for h_v: -0.407 and $+0.255$). The within-graph value is itself a mixture of two opposite mechanisms, one per perturbation:

- *Experiment A (uniform link removal) lowers h.* The mean h falls from 0.617 at 90% of arcs kept to 0.476 at 20% (within-graph correlation with the kept fraction $+0.721$). Removing arcs creates dangling pages, and their uniform rows add n_{d}/n to every column sum, pulling Σ_{in} towards 1 (correlation of h with the dangling fraction -0.923).
- *Experiment B (newly added pages) raises h*, from 0.644 to 0.704: pages that lose 90% of their in-links acquire $\Sigma_{\mathrm{in}} \approx 0$ (correlation of h with the share of zero-inflow pages $+0.502$).

In both experiments α^* *decreases* with severity, from 0.841 to 0.776 (A) and from 0.834 to 0.704 (B). The within-graph correlation of h with α^* is therefore $+0.446$ under A and -0.520 under B. A statistic that rises under one kind of link loss and falls under the other, while the optimal damping falls under both, cannot calibrate α. The initial hypothesis that a larger h calls for a larger α is contradicted, and so is its opposite. Table 6 quantifies this. Under NCD, h alone explains -0.02 of the variance of α^* out of sample, and at most 0.05 under any truth. Graph identity explains 0.25 and the perturbation level, which is not observable in practice, 0.47; boosted trees on the hidden generator parameters plus the level recover 0.60 out of sample, whereas every observable statistic together recovers only 0.17.

Experiments A and B are mixtures of dangling-page creation, zero-in-link pages and altered transition mass, with different proportions. The generated table makes this composition explicit. In particular, h_{raw} alone is nearly blind to A, so it is not a general-purpose severity measure for both perturbation protocols.

Non-monotone and dangling-deconfounded rules. Because α^* falls under both perturbations while h moves in opposite directions, within one graph α^* is an inverted V in h around the graph's unperturbed value h_0 (the joined series in Figure 4b). The distance from that apex carries real information: $|h - h_0|$ correlates -0.459 with α^* (-0.496 within graphs). We therefore searched for a rule that exploits the shape (script `stage2_nonmonotone_search.py`; Table 8). Asymmetric tents $\mathrm{clip}(\alpha_{\mathrm{peak}} - s_L \max(0, h_0 - h) - s_R \max(0, h - h_0),\ 0.40,\ 0.99)$, symmetric tents, a concave quadratic, a Gaussian bump, two- and three-region step functions

Table 6: Share of the variance of the oracle damping α^* explained (R^2) on the calibration panel, for each truth and r-setting. Graph identity, perturbation level and their additive combination are in-sample ANOVA shares. Starred columns are out of sample (grouped 8-fold CV by graph): boosted trees on the hidden generator parameters plus the perturbation level; ridge regression on every observable global statistic, including h and h_v; and the clip-linear rule on h alone. Negative values mean worse than predicting the mean.

truth	r	graph	level	graph+level	hidden+level*	all observables*	h alone*
NCD	Stage 1	0.25	0.47	0.72	0.60	0.17	−0.02
NCD	0.118	0.32	0.41	0.73	0.62	0.13	−0.03
NCD95	0.105	0.40	0.22	0.62	0.38	0.10	−0.03
TOT	Stage 1	0.10	0.77	0.87	0.87	0.18	0.05
POP	POP rule	0.54	0.29	0.83	0.44	0.28	−0.03
Q	Stage 1	0.67	0.02	0.69	−0.20	0.05	−0.03

Table 7: Experiments A and B are mixtures of dangling-page and zero-in-link effects. Values run from the least damaged to the most severe level.

experiment	δ	ζ	h	h_{raw}
A: link keep 1.0→0.2	0.173→0.483	0.156→0.506	0.620→0.476	0.679→0.669
B: page loss 0.1→0.6	0.185→0.298	0.207→0.461	0.644→0.704	0.707→0.827

and a two-knot linear spline were fitted on h and on h_v, both by least squares to α^* and by direct maximisation of mean τ (exactly for the step functions; otherwise by grid search, with the apex of rules on h in $[0.50, 0.70]$, followed by Nelder–Mead and Powell). They were evaluated exactly as the other candidates: 8-fold cross-validation by graph on `calib`, with the constant refitted in every fold, then the four held-out panels. The τ-optimised tent places its apex at $h_0 = 0.525$, with $\alpha_{\text{peak}} = 0.821$, $s_L = 0.244$ and $s_R = 0.053$. It does follow part of the within-graph variation (its predictions correlate +0.188 with α^* within graphs), yet it changes mean τ by only +0.0001 in sample, −0.0001 in cross-validation ($p = 0.538$) and −0.0001 on the 73 held-out graphs ($p = 0.290$). No τ-optimised non-monotone rule on h or h_v gains more than +0.0001 in cross-validation or +0.0002 out of sample, and the least-squares fits change τ by −0.0013 to −0.0017 out of sample.

Removing the uniform dangling contribution $(1 - r)\,\delta$, with $\delta = n_{\text{d}}/n$, from every column sum tests whether the fall of h under Experiment A is a dangling artefact. Largely it is: the deconfounded deviation $h_{\text{raw}} = \frac{1}{n}\sum_v |1 - \Sigma_{\text{in}}(v) + (1 - r)\delta|$ barely moves under A (0.679 unperturbed, 0.669 at 20% of arcs kept) and rises under B (0.827 at 60%). Two further variants rise under both perturbations: the relative dispersion $h^{\text{rel}}_{\text{raw}}$ of the deconfounded sums (from 0.723 to 0.759 under A and 0.919 under B) and the deviation h^{H}_{raw} of the column sums of the hyperlink matrix before its dangling rows are patched (from 0.801 to 0.838 and 1.013). Within graphs, both correlate negatively with α^* under each experiment (−0.594 and −0.827 for $h^{\text{rel}}_{\text{raw}}$). Deconfounding thus turns the inverted V into a monotone relation, but only inside each graph. The unperturbed level of $h^{\text{rel}}_{\text{raw}}$ varies across graphs with standard deviation 0.124, against within-graph excursions of +0.036 (A) and +0.196 (B), and it correlates *positively* with the graph-mean α^* (+0.376): the Simpson structure of Stage 1 again. The τ-optimised tents on the three deconfounded statistics degenerate to a decreasing hinge, and no rule on them gains more than +0.0003 in cross-validation or +0.0003 out of sample.

The two raw counts that rise under both perturbations give the strongest deployable rules: the dangling fraction δ (from 0.173 to 0.483 under A and 0.298 under B; Experiment B also creates dangling pages, because it deletes the out-arcs of pages whose links all pointed to the new pages) and the share ζ of pages without in-links (from 0.156 to 0.506 and 0.461). The bivariate rule $\text{clip}(0.918 - 0.101\,h - 0.179\,\delta)$ gains +0.0007 in cross-validation ($p = 0.001$) but +0.0004 out of sample ($p = 0.119$). The clipped-linear rule $\alpha = \text{clip}(0.849 - 0.140\,\zeta)$ gains +0.0014 in cross-validation ($p = 2.0 \times 10^{-7}$) and +0.0003 on the held-out graphs ($p = 0.039$); adding h leaves it unchanged (coefficient −0.003). Its gain comes from the two panels drawn from the calibration generator (+0.0009 on `test`, +0.0012 on `scale`), shrinks to +0.0004 on `topo` and reverses on `real` (−0.0060, worse on three of the five graphs). The unperturbed share of pages without in-links is 0.156 on `calib` but 0.391 on `topo` and 0.328 on `real` (0.40–0.63 on the three citation graphs), so its level, too, mixes the graph with the perturbation.

Table 8: Non-monotone, dangling-deconfounded and bivariate Stage 2 rules against the locked constant $\alpha = 0.810$ (truth NCD, r = Stage 1 rule), selected from the 127 deployable rules of `stage2_nonmonotone_search.py`. Entries are differences in mean Kendall τ in units of 10^{-4}, rule minus constant (the decision threshold 0.0005 is 5.0). calib: in sample; CV: grouped 8-fold cross-validation on `calib`, with the constant refitted in every fold; held-out: the 73 graphs of `test`, `scale`, `topo` and `real` pooled. p: Wilcoxon signed-rank test over per-graph means; †: Holm-adjusted $p < 0.05$ across the four panels. Rules are τ-optimised unless marked LS. h_0 and $h_{\mathrm{raw},0}$ are the values on the graph's own unperturbed version; the two ceilings are in sample.

rule	calib	CV	p_{CV}	held-out	p	`test`	`scale`	`topo`	`real`
non-monotone rules on h and h_v									
asymmetric tent, LS	−11.8	−14.5	3.9×10^{-4}	−16.6	2.6×10^{-7}	$-17.7^{\dagger}$	−17.5	$-7.2^{\dagger}$	−52.4
asymmetric tent	+0.9	−1.0	0.538	−1.1	0.290	−1.2	−3.3	−0.1	−0.6
symmetric tent	+0.3	−1.8	0.412	−0.7	0.477	+0.0	−2.2	−0.8	−0.7
concave quadratic	0.0	−1.9	0.072	0.0	–	0.0	0.0	0.0	0.0
Gaussian bump	+1.3	+0.1	0.915	+0.4	0.263	+0.5	−2.6	+0.1	+8.8
three-region step	+1.7	+0.5	0.586	−0.8	0.166	−1.5	−2.4	+0.2	+2.3
two-knot linear spline	+1.4	−0.8	0.963	−0.1	0.343	−0.3	−2.2	+0.1	+5.0
asymmetric tent on h_v	+0.0	−2.4	0.784	−0.2	0.245	−0.2	+0.1	−0.0	−1.7
dangling-deconfounded and severity statistics									
tent on h_{raw}	+1.0	−0.3	0.108	+1.0	0.245	$+3.0^{\dagger}$	−0.0	−1.3	+1.7
tent on $h_{\mathrm{raw}}^{\mathrm{rel}}$	+3.2	+1.6	0.009	+1.2	0.366	$+5.1^{\dagger}$	+1.6	−3.1	−4.9
tent on h_{raw}^{H}	+3.5	+2.1	0.008	+0.9	0.562	$+5.2^{\dagger}$	+1.7	$-0.0^{\dagger}$	−24.7
two-region step on δ	+4.8	+4.5	2.0×10^{-5}	+3.0	0.005	$+3.8^{\dagger}$	+7.5	−1.3	+8.2
clipped-linear on ζ	+15.0	+14.2	2.0×10^{-7}	+3.1	0.039	$+9.0^{\dagger}$	+12.4	+3.8	−59.6
three-region step on ζ	+13.7	+10.3	3.9×10^{-5}	+4.0	0.014	$+8.2^{\dagger}$	+12.0	+0.7	−26.4
bivariate rules									
clipped-linear in (h, δ)	+8.4	+6.6	0.001	+3.5	0.119	+6.4	+12.0	−3.3	−2.4
tent in $h + c\,\delta$	+8.3	+6.4	0.002	+3.2	0.155	+5.8	+11.0	−3.4	−0.1
3×3 cells in (h, δ)	+9.0	+6.6	2.5×10^{-4}	+1.5	0.297	$+6.3^{\dagger}$	+3.9	−4.8	−5.7
boosted trees on (h, δ), LS	+16.7	−21.2	0.101	−22.8	3.3×10^{-4}	−10.8	−16.4	$-20.9^{\dagger}$	−125.0
clipped-linear in (h, ζ)	+15.0	+13.7	7.2×10^{-7}	+3.3	0.046	+9.0	+12.3	+4.4	−60.1
not deployable: need the unperturbed graph or the truth									
tent in $h - h_0$	+19.2	+17.8	7.0×10^{-7}	+5.7	0.018	+9.7	+11.0	+2.7	−18.4
three-region step in $h_{\mathrm{raw}} - h_{\mathrm{raw},0}$	+17.3	+15.5	4.3×10^{-7}	+9.7	6.1×10^{-5}	$+11.6^{\dagger}$	+13.3	+3.5	+18.9
best constant per graph	+9.1	–	–	+15.4	–	+8.2	+10.8	+11.3	+92.3
oracle α^*	+52.2	–	–	+53.3	–	+48.6	+52.1	+35.6	+171.9

The obstacle is the location of the apex, not the shape of g. Given each graph's own unperturbed h_0, which the observed graph does not reveal, a τ-optimised tent in $h - h_0$ gains +0.0018 in cross-validation ($p = 7.0 \times 10^{-7}$; 34% of the oracle headroom of +0.0052) and +0.0006 out of sample ($p = 0.018$). A step rule in $h_{\mathrm{raw}} - h_{\mathrm{raw},0}$ gains +0.0010 out of sample ($p = 6.1 \times 10^{-5}$), on all four held-out panels. With one apex for every graph, the correlation of α^* with the distance falls from −0.459 to −0.133 (best common apex 0.55): the apex varies across graphs with standard deviation 0.114, as much as h moves within a graph under the perturbations (mean excursions −0.144 under A and +0.084 under B). None of the 127 deployable rules meets the decision rule fixed before the held-out evaluation: a gain of at least 0.0005 with $p < 0.05$ both in cross-validation and on the pooled held-out panels, and no Holm-significant loss on any panel. The cross-validation part is passed by 21 rules, the held-out part by none. The constant $\alpha = 0.810$ therefore remains the Stage 2 rule. The same search under NCD95 (constant 0.930) and TOT (constant 0.480) promotes no rule either, while the tent in $h - h_0$ is again significant out of sample (+0.0004, $p = 1.3 \times 10^{-4}$, and +0.0020, $p = 3.1 \times 10^{-7}$).

8.3 Anchoring and the noise discount

Table 9 and Figure 5(a, b) show the effect that dominates Stage 2: the oracle damping is set by the reference ranking. On the unperturbed graph, the oracle equals the damping that defines the truth: 0.852 for NCD (0.85) and 0.950 for NCD95 (0.95). It is 0.547 for TOT, whose Beta(1,1) prior has mean 0.5, 0.767 for POP and 0.971 for the walk-free Q. Under perturbation it sits slightly lower: 0.791, 0.914, 0.471, 0.726 and 0.968 respectively. POP is the only truth whose anchor lies clearly below its own damping even on the unperturbed

Table 9: Oracle damping under the five truths (calibration panel). Own α: the damping that defines the truth ([a]TotalRank: mean of the Beta(1,1) prior; [b]POP is the PageRank of the expected walk at 0.85). α^* (A1.0): mean oracle on the unperturbed graphs. Mean α^*: mean over the perturbed instances; discount: their difference. Const α: the best constant on the calibration panel. Within/between: correlation of h with α^* inside graphs and between graph means.

truth	r	own α	α^* (A1.0)	mean α^*	discount	const α	within	between
NCD	Stage 1	0.85	0.852	0.791	−0.061	0.81	−0.30	+0.29
NCD95	0.105	0.95	0.950	0.914	−0.036	0.93	−0.22	+0.26
TOT	Stage 1	0.50[a]	0.547	0.471	−0.075	0.48	−0.37	−0.15
POP	POP rule	0.85[b]	0.767	0.726	−0.042	0.74	−0.54	+0.14
Q	Stage 1	–	0.971	0.968	−0.003	0.99	−0.19	+0.18

Table 10: Dose-response test of anchoring: oracle damping α^* against Random-α NCDawareRank references $\mathbb{E}[\pi(A)]$ with $A \sim \text{Beta}(a, b)$, on 16 calibration graphs (perturbed instances only, r = Stage 1 rule, grid 0.05–0.99). Beta(1,1) is TotalRank.

prior	prior mean	mean α^*	s.d.	α^*−mean
Beta(3,7)	0.30	0.267	0.046	−0.033
Beta(5,5)	0.50	0.447	0.060	−0.053
Beta(1,1)	0.50	0.460	0.062	−0.040
Beta(7,3)	0.70	0.636	0.063	−0.064
Beta(8.5,1.5)	0.85	0.793	0.056	−0.057

graph. This is expected: POP is defined on the generator's expected walk, and every realised graph is already a noisy copy of that walk before any link is removed. The two α-free references disagree completely, TOT pointing to $\alpha \approx 0.5$ and Q to the top of the grid. For Q the optimum lies at the top of the grid rather than beyond it: on 8 graphs, mean τ against q is 0.4282, 0.4279 and 0.4273 at $\alpha = 0.99$, 0.995 and 0.999.

The dose-response test in Table 10 isolates the mechanism. We vary the prior of Random-α NCDawareRank references and nothing else. The mean oracle damping then follows the prior mean with slope 0.950 (intercept −0.021): 0.267, 0.447, 0.636 and 0.793 for prior means 0.30, 0.50, 0.70 and 0.85. The offset of −0.03 to −0.06 is the *noise discount*: when the observed graph is a perturbed copy of the reference graph, the best single damping is slightly lower than the reference's own. The natural interpretation is that long walks propagate the noise introduced by missing links, so a slightly shorter effective walk recovers the reference better. Stage 2's move from 0.85 to 0.810 under NCD is exactly this effect. Under NCD95 the corresponding constant is 0.930, which improves on $\alpha = 0.95$ by +0.0011 on `test` ($p = 0.062$), +0.0015 on `scale`, +0.0026 on `topo` ($p = 1.3 \times 10^{-4}$) and −0.0006 on `real`.

The anchor-matched operational claim is conditional on the reference: use $\alpha = 0.81$ when the reference is defined at 0.85, and $\alpha = 0.93$ when it is defined at 0.95. The first changes the NCD comparison by +0.0110, while the matched NCD95 comparison changes by +0.0011 ($p = 0.062$), a tie on `test`. The reference damping is a methodological decision, and Table 11 shows what a mismatch costs on `test`: under the NCD truth, a user whose reference is the canonical (0.118, 0.85) and who adopts the 0.95-anchored constant changes τ by −0.0441; under the NCD95 truth, a user whose reference is (0.105, 0.95) and who adopts $\alpha = 0.81$ changes it by −0.0303. With the Stage 1 rule in place of $r = 0.105$, the same constant 0.93 improves on the [NG13] setting (0.105, 0.95) by +0.0089 on `test` (30/32 graphs, $p = 5.0 \times 10^{-7}$); the best constant for that pairing on `calib` is 0.920, and its τ exceeds that of 0.93 by only +0.0002. The two-stage pipeline also beats the [NG13] ranking setting (0.105, 0.95) by +0.0743 under the NCD truth, but that number mostly reflects the fact that the NCD truth is itself defined at the damping 0.85; it is not evidence about the method and is not used in any claim.

Practical recipe. Choose the reference anchor first. For 0.85, call `tune(..., alpha_ref=0.85)`: it returns Stage 1 r and $\alpha = 0.81$. For 0.95, use `alpha_ref=0.95`: it returns the same Stage 1 r and $\alpha = 0.93$. The calibration optimum for this pairing is 0.920, only +0.0002 τ above 0.93. The tool does not return the reference's own $r = 0.10/\alpha_{\text{ref}}$; it returns Stage 1 r by design. A baseline argument is ignored with a warning because no baseline rule is validated.

Table 11: Anchor matching. Mean Kendall τ difference (candidate minus comparator), averaged over graphs, with graph wins in parentheses, on the four held-out panels; † marks an exact Wilcoxon $p \geq 0.05$ (every other cell has $p \leq 0.0269$). The comparators are the static [NG13] settings (r, α). Cells marked * are the anchor-matched comparisons claimed in the text; the others are deliberate mismatches, in which the candidate, the comparator or the truth is defined at the other damping. With five real graphs the smallest attainable p is 0.0625.

candidate (r, α) vs comparator	`test`	`scale`	`topo`	`real`
truth NCD				
rule r, $\alpha = 0.81$ vs (0.118, 0.85)*	+0.0110 (31/32)	+0.0102 (11/12)	+0.0036 (23/24)	+0.0017 (2/5)†
rule r, $\alpha = 0.81$ vs (0.105, 0.95)	+0.0743 (32/32)	+0.0756 (12/12)	+0.0515 (24/24)	+0.0848 (5/5)†
rule r, $\alpha = 0.93$ vs (0.118, 0.85)	−0.0414 (0/32)	−0.0472 (0/12)	−0.0318 (0/24)	−0.0777 (0/5)†
rule r, $\alpha = 0.93$ vs (0.105, 0.95)	+0.0219 (31/32)	+0.0182 (11/12)	+0.0162 (24/24)	+0.0053 (4/5)†
$r = 0.105$, $\alpha = 0.93$ vs (0.118, 0.85)	−0.0441 (0/32)	−0.0455 (0/12)	−0.0300 (0/24)	−0.0528 (0/5)†
$r = 0.105$, $\alpha = 0.93$ vs (0.105, 0.95)	+0.0192 (32/32)	+0.0199 (12/12)	+0.0180 (24/24)	+0.0303 (5/5)†
truth NCD95				
rule r, $\alpha = 0.81$ vs (0.118, 0.85)	−0.0109 (6/32)	−0.0124 (2/12)	−0.0069 (2/24)	−0.0206 (0/5)†
rule r, $\alpha = 0.81$ vs (0.105, 0.95)	−0.0303 (2/32)	−0.0306 (1/12)	−0.0172 (1/24)	−0.0654 (0/5)†
rule r, $\alpha = 0.93$ vs (0.118, 0.85)	+0.0283 (31/32)	+0.0245 (10/12)	+0.0137 (21/24)	+0.0424 (5/5)†
rule r, $\alpha = 0.93$ vs (0.105, 0.95)*	+0.0089 (30/32)	+0.0062 (10/12)	+0.0034 (20/24)	−0.0025 (1/5)†
$r = 0.105$, $\alpha = 0.93$ vs (0.118, 0.85)	+0.0205 (32/32)	+0.0197 (11/12)	+0.0130 (23/24)	+0.0443 (5/5)†
$r = 0.105$, $\alpha = 0.93$ vs (0.105, 0.95)*	+0.0011 (19/32)†	+0.0015 (7/12)†	+0.0026 (20/24)	−0.0006 (2/5)†

8.4 Complete pipelines against the static settings

Table 12 compares complete (r, α) pipelines with the two published static settings, under all five truths and on all panels. Under the primary truth, the two-stage pipeline (locked rule for r, $\alpha = 0.810$) improves on the canonical setting $(0.118, 0.85)$ by +0.0110 on `test` (95% CI $[0.0089, 0.0133]$, better on 31/32 graphs, $p = 9.3 \times 10^{-10}$), +0.0102 on `scale` (11/12) and +0.0036 on `topo` (23/24). This is about two thirds of the oracle headroom on `test` (+0.0159). The gain splits into +0.0066 from Stage 1 ($p = 2.0 \times 10^{-6}$) and +0.0044 from Stage 2 ($p = 1.3 \times 10^{-6}$). On `real` it is +0.0017 per graph (95% CI $[-0.0037, 0.0097]$, 2/5 graphs, $p = 0.812$): +0.0172 on cnr-2000, +0.0020 on Cora, −0.0024 on CiteSeer, −0.0032 on Cora-ML and −0.0049 on WikiCS (Appendix Table 19).

The verdict depends on the truth (Figure 5d). The same locked pipeline:

- beats the canonical setting under TOT (+0.0422) and POP (+0.0139), simply because 0.81 is closer than 0.85 to those truths' anchors;
- loses to the setting $(0.105, 0.95)$ of [NG13] under NCD95 (−0.0303) and under the walk-free Q (−0.0455), whose anchors lie above 0.9.

Claims that a calibrated damping "beats the default" are therefore claims about the reference as much as about the method.

8.5 Convergence

Lowering α also shortens the power iteration. This holds for any lower α and is a side benefit, not a property of the calibration. The a-priori estimate $k \approx \ln \varepsilon / \ln \alpha$, based on $|\lambda_2(P)| \leq \alpha$ [HK03, NG13], predicts savings of 22.9% for $0.85 \to 0.81$ ($85 \to 66$ iterations at $\varepsilon = 10^{-6}$) and 29.3% for $0.95 \to 0.93$. The measured savings are smaller. The actual contraction is faster than α, so both iteration counts stay well below the estimate (Table 13), and the saving is 14.7% on `test` and 13.1% on `real` for the NCD branch, and 19.3% and 15.3% for the NCD95 branch.

8.6 Sensitivity truths with refitted rules

Refitting g separately for each truth does not change the conclusion (Appendix Table 18). Under Q, POP and NCD95, the τ-optimised rule on h is the constant or indistinguishable from it. Under TOT, h carries a weak signal with the same negative sign within and between graphs. It gains +0.0006 on `test` ($p = 0.002$), but −0.0001 on `topo` and −0.0020 on `real`, so it does not transfer.

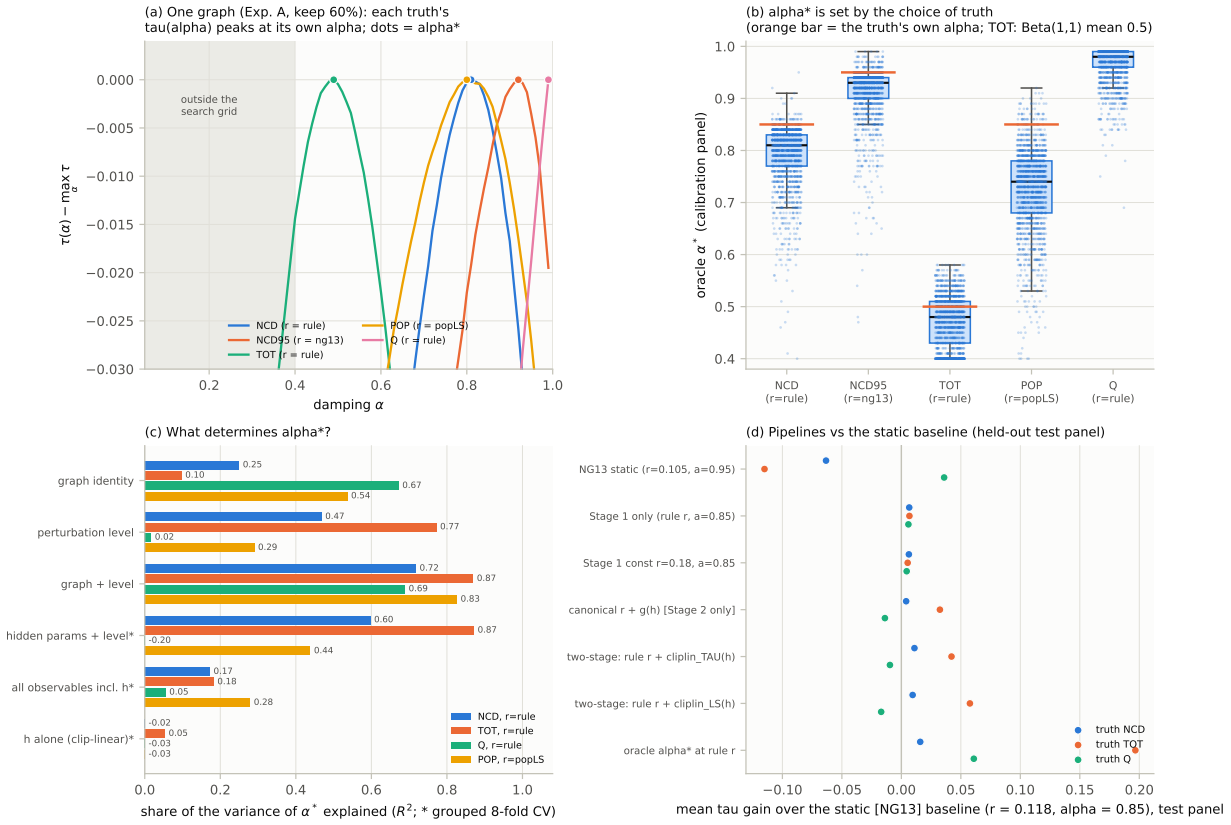


Figure 5: Stage 2 diagnostics. **(a)** For one calibration graph (Experiment A, 60% of arcs kept), the curve $\tau(\alpha) - \max \tau$ under each truth peaks near the damping that defines that truth. **(b)** Distribution of α^* per truth; orange bars mark each truth's own α. **(c)** Share of the variance of α^* explained. **(d)** Gain over the static setting $(0.118, 0.85)$ on `test` for complete pipelines under three truths. The sign of the two-stage gain flips between truths.

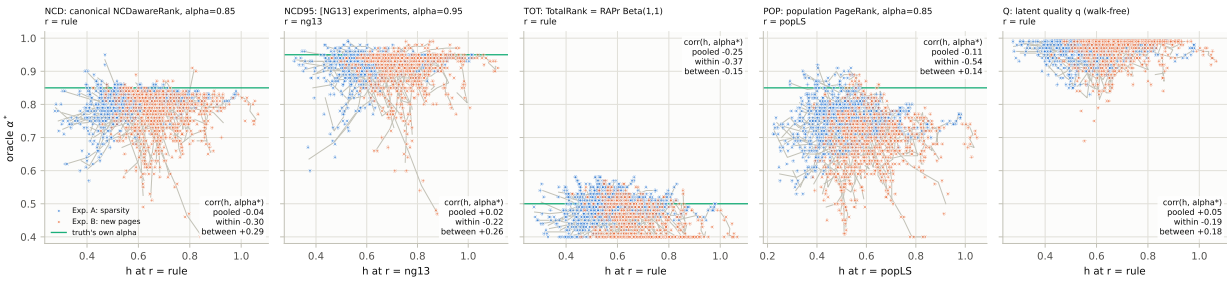


Figure 6: Oracle damping against h under the five truths (calibration panel). Within each truth the relation with h is weak and of mixed sign; across truths the level of α^* follows the damping that defines the truth (green line).

Table 12: Complete pipelines against the static [NG13] settings under all five truths. Cells: mean Kendall τ over instances; in parentheses, the difference against [NG13] (0.118, 0.85) averaged over graphs (each graph weighted equally). *Two-stage*: r from the locked Stage 1 rule and $\alpha = 0.81$, both fitted on calibration graphs under NCD and applied unchanged under every truth. POP and Q exist only for the static generator.

policy (r, α)	`calib`	`test`	`scale`	`topo`	`real`
truth NCD					
[NG13] (0.118, 0.85)	0.6805	0.6801	0.6633	0.7840	0.7559
[NG13] (0.105, 0.95)	0.6133 (−0.0673)	0.6168 (−0.0633)	0.5979 (−0.0654)	0.7360 (−0.0479)	0.6792 (−0.0831)
Stage 1 only	0.6862 (+0.0056)	0.6867 (+0.0066)	0.6681 (+0.0049)	0.7844 (+0.0005)	0.7600 (+0.0030)
two-stage ($\alpha = 0.81$)	0.6913 (+0.0108)	0.6911 (+0.0110)	0.6735 (+0.0102)	0.7875 (+0.0036)	0.7559 (+0.0017)
oracle α^*	0.6965 (+0.0160)	0.6960 (+0.0159)	0.6787 (+0.0155)	0.7911 (+0.0071)	0.7738 (+0.0189)
truth NCD95					
[NG13] (0.118, 0.85)	0.6227	0.6237	0.6044	0.7472	0.6678
[NG13] (0.105, 0.95)	0.6410 (+0.0182)	0.6431 (+0.0194)	0.6226 (+0.0183)	0.7576 (+0.0104)	0.7122 (+0.0449)
Stage 1 only	0.6282 (+0.0055)	0.6306 (+0.0069)	0.6096 (+0.0052)	0.7472 (−0.0001)	0.6767 (+0.0104)
two-stage ($\alpha = 0.81$)	0.6103 (−0.0125)	0.6128 (−0.0109)	0.5920 (−0.0124)	0.7404 (−0.0069)	0.6463 (−0.0206)
oracle α^*	0.6530 (+0.0303)	0.6560 (+0.0323)	0.6343 (+0.0299)	0.7660 (+0.0188)	0.7281 (+0.0609)
truth TOT					
[NG13] (0.118, 0.85)	0.5893	0.5932	0.5773	0.7624	0.6804
[NG13] (0.105, 0.95)	0.4689 (−0.1204)	0.4782 (−0.1150)	0.4591 (−0.1183)	0.7001 (−0.0623)	0.5387 (−0.1461)
Stage 1 only	0.5963 (+0.0071)	0.6000 (+0.0068)	0.5854 (+0.0080)	0.7655 (+0.0030)	0.6688 (−0.0128)
two-stage ($\alpha = 0.81$)	0.6341 (+0.0449)	0.6354 (+0.0422)	0.6230 (+0.0457)	0.7775 (+0.0151)	0.7097 (+0.0298)
oracle α^*	0.7987 (+0.2094)	0.7899 (+0.1967)	0.7839 (+0.2065)	0.8094 (+0.0469)	0.8336 (+0.1582)
truth POP					
[NG13] (0.118, 0.85)	0.5462	0.5446	0.5172	–	–
[NG13] (0.105, 0.95)	0.4904 (−0.0558)	0.4957 (−0.0490)	0.4683 (−0.0489)	–	–
Stage 1 only	0.5510 (+0.0048)	0.5506 (+0.0060)	0.5237 (+0.0065)	–	–
two-stage ($\alpha = 0.81$)	0.5614 (+0.0152)	0.5585 (+0.0139)	0.5326 (+0.0154)	–	–
oracle α^*	0.5745 (+0.0283)	0.5677 (+0.0231)	0.5443 (+0.0271)	–	–
truth Q					
[NG13] (0.118, 0.85)	0.3655	0.3478	0.3302	–	–
[NG13] (0.105, 0.95)	0.3939 (+0.0283)	0.3837 (+0.0359)	0.3560 (+0.0258)	–	–
Stage 1 only	0.3724 (+0.0069)	0.3537 (+0.0059)	0.3332 (+0.0030)	–	–
two-stage ($\alpha = 0.81$)	0.3585 (−0.0070)	0.3382 (−0.0096)	0.3202 (−0.0100)	–	–
oracle α^*	0.4150 (+0.0495)	0.4087 (+0.0609)	0.3712 (+0.0410)	–	–

Table 13: Power-iteration counts before and after the noise discount (stopping rule $\|\pi^{(k+1)} - \pi^{(k)}\|_1 < \varepsilon = 10^{-6}$, start v). The bound column is the a-priori estimate from $|\lambda_2(P)| \le \alpha$; measured columns are mean counts over all perturbed instances of the panel.

configuration	α	bound $\ln \varepsilon / \ln \alpha$	measured, `test`	measured, `real`
NCD branch (r = rule)	0.85→0.81	85→66 (−22.9%)	30.2→25.8 (−14.7%)	30.5→26.5 (−13.1%)
NCD95 branch ($r = 0.105$)	0.95→0.93	269→190 (−29.3%)	67.2→54.2 (−19.3%)	69.9→59.2 (−15.3%)

9 V2 extension: external real networks and operational checks

The preregistered pooled real2 analysis combines the five old real graphs with the six new graphs, using graphs rather than perturbations as the replication units ($n = 11$). H1 is −0.0021 (4/11, $p = 0.465$) and H3b is −0.0037 (2/11, $p = 0.102$). H2 is +0.0760 (11/11, $p = 9.8 \times 10^{-4}$), but it compares different reference anchors and is descriptive only. The six-new-graph Holm family is untestable under Amendment 1. Cornell is influential (183 nodes, 295 arcs, 48% dangling; H1 gain −0.0355). The labels are proxies for topical or department structure, and the SNAP partitions are host proxies, so generalisation of the gain to new real networks is not supported.

The fitted operational anchor discount is not promoted. It gains −0.0008 on `test`, −0.0004 on `topo`, and +0.0002 on `real`; the topology loss is significant and the pooled held-out result fails the preregistered threshold.

The graph-unit mixed-damage contrast does not promote C4–C7 over C1. The reduced B1 baseline smoke run remains descriptive only (mean gain +0.0051; B2 and B3 were not run).

Table 14: Pooled real2 per-graph means for H1, H2 and H3b. H2 is anchor-mismatched and descriptive; the pooled inferential analysis has 11 graphs.

graph	cohort	partition	H1	H2	H3b
citeseer	old	label	-0.0024	+0.0636	-0.0092
cnr2000	old	proxy	+0.0172	+0.1578	+0.0072
cora_full	old	label	+0.0020	+0.0923	-0.0008
cora_ml	old	label	-0.0032	+0.0554	-0.0029
wikics	old	label	-0.0049	+0.0550	-0.0067
cornell	new	label	-0.0355	+0.0699	-0.0061
email-Eu-core	new	label	-0.0052	+0.0566	-0.0067
film	new	label	-0.0012	+0.0238	-0.0005
ogbn-arxiv	new	label	+0.0128	+0.1826	+0.0030
texas	new	label	+0.0012	+0.0439	-0.0156
wisconsin	new	label	-0.0044	+0.0353	-0.0026

Table 15: Formal decision-rule results for the fitted anchor discount versus no discount.

comparison	mean gain	wins/n	exact p	bootstrap 95% CI
calib CV	+0.0031	45/64	0.0000	[+0.0020, +0.0042]
pooled held-out	-0.0005	22/61	0.0646	[-0.0018, +0.0008]
test	-0.0008	14/32	0.3217	[-0.0026, +0.0013]
topo	-0.0004	6/24	0.0340	[-0.0013, +0.0007]
real	+0.0002	2/5	1.0000	[-0.0086, +0.0091]

Table 16: Graph-unit pooled held-out mixed-damage contrast versus C1; no rule is promoted.

rule	graphs	mean vs C1	wins	exact p
C4	36	-0.0003	22	0.3074
C5	36	-0.0024	15	0.1079
C6	36	+0.0002	13	0.0395
C7	36	-0.0012	13	0.0334

10 Discussion

What the calibration achieves. Against the canonical reference, the two-stage procedure amounts to retuning two constants: more weight on the proximity matrix ($r \approx 0.2$, $\mu \approx 0.17$ instead of 0.10) and slightly less damping ($\alpha = 0.810$ instead of 0.85). On held-out synthetic graphs, including larger graphs and a different generative mechanism, the gain is 0.004–0.011 in Kendall's τ, consistent across graphs, plus a modest acceleration. On the pooled old-plus-new real networks, H1 is near zero and H3b is slightly negative; generalisation of the gain to new real networks is not supported. These are honest, if modest, results for a zero-cost procedure. They are not the adaptive, structure-driven calibration that motivated the study.

Why structural statistics fail here. The oracle parameters depend on two quantities: how severely the observed graph is perturbed relative to the reference, and the latent quality structure of the pages. Neither is revealed by global statistics of the observed graph. $\bar{P}_{\text{esc}}^{(M)}$ and h react strongly to link loss inside a graph. Across graphs their levels are confounded with ordinary structural variation, and h responds to different kinds of link loss with opposite signs. Section 8.2 isolates the missing ingredient: given the unperturbed graph's own value h_0, a tent in $h - h_0$ recovers 34% of the oracle headroom in cross-validation, whereas none of 127 rules on statistics of the observed graph alone passes the held-out test. The statistic could therefore become useful where a less perturbed snapshot of the same graph is available as a reference; we have not tested this. A related pattern appeared in our earlier work on uncertain graphs [Kon26]: there, the correlation between the escape probability and the optimal damping did not survive on real networks, and routing among heuristics by the same statistic worked on synthetic graphs but failed on real ones. We take this as evidence that single global statistics are too coarse to transfer parameter choices between graphs.

Anchoring as a methodological lesson. The strongest finding is not about NCDawareRank. When a ranking method is tuned to agree with a reference ranking, the optimal damping is the reference's own damping minus a noise discount. This follows from the dose-response test and from five truths with very different anchors. Any statement that one damping factor is "better" therefore inherits the damping of the ground truth. The two α-free references available to us disagree by almost 0.5, which suggests that no truth-independent optimal damping exists for this problem. Evaluations of damping choices should report results under several references, and stability protocols should not be used to choose parameters.

What the theory of h offers. The identities of Proposition 3 and the bound of Theorem 1 give h and h_v a precise meaning: they measure how strongly the chain pushes rank away from the uniform and the teleportation distribution, and h_v bounds the total displacement of the ranking for every α. They are useful diagnostics, for example for deciding whether the damping matters at all for a given graph. The empirical results show that they are not calibration statistics.

Limitations. The synthetic generators, however varied, are our own. The partition of cnr-2000 is a host proxy, the citation graphs use topic labels, and the new labels and SNAP partitions are proxies rather than external relevance judgements. Cornell materially affects H1. Dangling pages are handled by uniform rows, not by the block-confined strategy of [Nik20]. Stage 1 is anchored at $\alpha_0 = 0.85$, and the Stage 2 grid stops at 0.99, which censors the oracle under Q (we checked that the optimum does not lie beyond it). We evaluate rank correlations only, against references derived from models. External relevance judgements or browsing data, as in [GCFG10], would provide a ground truth that does not depend on a damping factor.

11 Conclusion

We formulated a two-stage, zero-cost calibration of NCDawareRank. The mixing ratio comes from the block escape probability and the total damping from the deviation of the ensembled operator from double stochasticity. We proved identities and a displacement bound that give these statistics a precise meaning. A large,

cluster-aware evaluation shows that the statistics do not predict the optimal parameters. The calibration reduces to two constants, $r \approx 0.2$ and $\alpha = 0.810$, that improve the recovery of the canonical ranking on synthetic graphs by 0.004–0.011 in Kendall's τ and do not generalise as a gain to the pooled real networks. The optimal damping is anchored to the damping of the reference ranking, minus a noise discount. The choice of ground truth therefore decides which parameter setting wins, and we recommend that future comparisons of link-analysis parameters report results under several references.

Reproducibility

All code, raw results and figures accompany the paper; the accompanying documentation describes every module. In brief:

- `prepare_real_graphs.py` builds the real panel;
- `run_experiments.py stage1` and `run_experiments.py stage2` run both stages;
- `stage1_analysis.py`, `stage2_analysis.py` and the decomposition, anchoring, convergence and bound-check scripts produce every number;
- `make_paper_assets.py` writes the numbers and tables of this manuscript directly from the result files.

Unit tests check the implementation against the matrices of [NG13, Fig. 1], Theorems 1–2 of [NG13], dense solves, quadrature and the bound (7).

References

[ALP08] K. Avrachenkov, N. Litvak, K. S. Pham. A singular perturbation approach for choosing the PageRank damping factor. *Internet Mathematics*, 5(1–2):47–69, 2008.

[BBC06] R. Baeza-Yates, P. Boldi, C. Castillo. Generic damping functions for propagating importance in link-based ranking. *Internet Mathematics*, 3(4):445–478, 2006.

[BCSV04] P. Boldi, B. Codenotti, M. Santini, S. Vigna. UbiCrawler: a scalable fully distributed Web crawler. *Software: Practice and Experience*, 34(8):711–726, 2004.

[BG18] A. Bojchevski, S. Günnemann. Deep Gaussian embedding of graphs: unsupervised inductive learning via ranking. In *Proc. ICLR*, 2018.

[BNG19] D. Berberidis, A. N. Nikolakopoulos, G. B. Giannakis. Adaptive diffusions for scalable learning over graphs. *IEEE Trans. Signal Processing*, 67(5):1307–1321, 2019.

[Bol05] P. Boldi. TotalRank: ranking without damping. In *Special Interest Tracks and Posters of the 14th WWW*, 898–899, 2005.

[BP10] M. Bressan, E. Peserico. Choose the damping, choose the ranking? *J. Discrete Algorithms*, 8(2):199–213, 2010.

[BSV05] P. Boldi, M. Santini, S. Vigna. PageRank as a function of the damping factor. In *Proc. 14th WWW*, 557–566, 2005.

[BSV09] P. Boldi, M. Santini, S. Vigna. PageRank: functional dependencies. *ACM Trans. Information Systems*, 27(4):19, 2009.

[BV04] P. Boldi, S. Vigna. The WebGraph framework I: compression techniques. In *Proc. 13th WWW*, 595–602, 2004.

[CG09] P. G. Constantine, D. F. Gleich. Random alpha PageRank. *Internet Mathematics*, 6(2):189–236, 2009.

[Cou77] P.-J. Courtois. *Decomposability: Queueing and Computer System Applications*. Academic Press, 1977.

[ET93] B. Efron, R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

[FBFM08] S. Fortunato, M. Boguñá, A. Flammini, F. Menczer. Approximating PageRank from in-degree. In *Algorithms and Models for the Web-Graph (WAW 2006)*, LNCS 4936, 59–71, 2008.

[Fri01] J. H. Friedman. Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

[GCFG10] D. F. Gleich, P. G. Constantine, A. D. Flaxman, A. Gunawardana. Tracking the random surfer: empirically measured teleportation parameters in PageRank. In *Proc. 19th WWW*, 381–390, 2010.

[HK03] T. H. Haveliwala, S. D. Kamvar. The second eigenvalue of the Google matrix. Technical Report 2003-20, Stanford InfoLab, 2003.

[Ken38] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938.

[Ken45] M. G. Kendall. The treatment of ties in ranking problems. *Biometrika*, 33(3):239–251, 1945.

[KH03] S. D. Kamvar, T. H. Haveliwala. The condition number of the PageRank problem. Technical Report 2003-36, Stanford InfoLab, 2003.

[Kon26] I. Konstantellos. Damping from the escape probability: a free calibration rule for diffusion-based reachability ranking in uncertain graphs. Computer Engineering and Informatics Department, University of Patras, 2026. Zenodo record `https://zenodo.org/records/23211440`.

[LM06] A. N. Langville, C. D. Meyer. *Google's PageRank and Beyond: The Science of Search Engine Rankings*. Princeton University Press, 2006.

[LSV07] N. Litvak, W. R. W. Scheinhardt, Y. Volkovich. In-degree and PageRank: why do they follow similar power laws? *Internet Mathematics*, 4(2–3):175–198, 2007.

[MC20] P. Mernyei, C. Cangea. Wiki-CS: a Wikipedia-based benchmark for graph neural networks. arXiv:2007.02901, 2020.

[NG13] A. N. Nikolakopoulos, J. D. Garofalakis. NCDawareRank: a novel ranking method that exploits the decomposable structure of the Web. In *Proc. 6th ACM WSDM*, 143–152, 2013.

[Nik20] A. N. Nikolakopoulos. Random surfing revisited: generalizing PageRank's teleportation model. arXiv:2008.12916, 2020.

[PBMW99] L. Page, S. Brin, R. Motwani, T. Winograd. The PageRank citation ranking: bringing order to the Web. Technical Report 1999-66, Stanford InfoLab, 1999.

[SA61] H. A. Simon, A. Ando. Aggregation of variables in dynamic systems. *Econometrica*, 29(2):111–138, 1961.

[Wil45] F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

A Additional tables

Table 17 repeats the Stage 1 comparison under the four Stage 1 truths. Table 18 refits the Stage 2 rule separately for each sensitivity truth. Table 19 breaks the real panel down by network.

Table 17: Stage 1 under the four Stage 1 truths. Const r: the best constant on the calibration panel; τ-opt. rule: the τ-optimised clip-linear rule on $\bar{P} = \bar{P}^{(M)}_{\text{esc}}$. Canonical τ: mean Kendall τ of the canonical $r = 0.118$ on `test`; const − can.: gain of the constant over it. The last four columns give the mean τ difference between the rule and the constant per panel. †: Wilcoxon signed-rank $p < 0.01$ over graphs. POP exists only for the static generator. Under SELF both the constant and the rule sit at the upper bound $r = 0.99$.

			`test`		rule − constant			
truth	const r	τ-opt. rule	canonical τ	const − can.	`test`	`scale`	`topo`	`real`
NCD	0.18	clip(0.242 − 0.143 $\bar{P}$)	0.6801	+0.0063†	+0.0003	0.0000	+0.0007	+0.0008
PR	0.02	clip(−0.017 − 0.075 $\bar{P}$)	0.6569	+0.0083	−0.0002	0.0000	+0.0016†	+0.0015
POP	0.23	clip(0.448 − 0.547 $\bar{P}$)	0.5446	+0.0067†	+0.0006	+0.0016	–	–
SELF	0.99	clip(0.960 + 0.836 $\bar{P}$)	0.6801	+0.1504†	0.0000	0.0000	0.0000	+0.0040

Table 18: Sensitivity truths, with g refitted for each truth: mean τ difference of the τ-optimised clip-linear rule on h against the truth's best constant α (Wilcoxon p over graphs). A dash means the panel has no such truth, or the rule equals the constant.

truth	const α	τ-optimised $g(h)$	`test`	`scale`	`topo`	`real`
TOT	0.48	clip(0.561 − 0.129 h)	+0.0006 (0.002)	+0.0002 (0.569)	−0.0001 (0.053)	−0.0020 (0.062)
Q	0.99	clip(0.939 + 0.093 h)	−0.0011 (0.008)	−0.0006 (0.359)	–	–
POP (r: LS rule)	0.74	clip(0.779 − 0.075 h)	+0.0003 (0.062)	−0.0002 (0.339)	–	–
POP (r: τ-opt. rule)	0.74	clip(0.761 − 0.041 h)	+0.0001 (0.098)	−0.0001 (0.339)	–	–
NCD95	0.93	clip(0.930 − 0.000 h)	0.0000 (–)	0.0000 (–)	0.0000 (–)	0.0000 (–)

Table 19: Real networks, truth NCD: mean Kendall τ per graph over the perturbed instances, and the mean oracle damping under NCD and TOT.

graph	mean r	[NG13] (0.118, 0.85)	Stage 1	two-stage	oracle	α^* NCD	α^* TOT
cnr-2000	0.230	0.7505	0.7435	0.7678	0.7788	0.772	0.521
WikiCS	0.181	0.7927	0.7933	0.7878	0.7956	0.835	0.508
Cora (full)	0.207	0.7513	0.7461	0.7533	0.7550	0.811	0.474
Cora-ML	0.225	0.7452	0.7392	0.7420	0.7471	0.831	0.480
CiteSeer	0.240	0.7370	0.7696	0.7346	0.7949	0.850	0.510